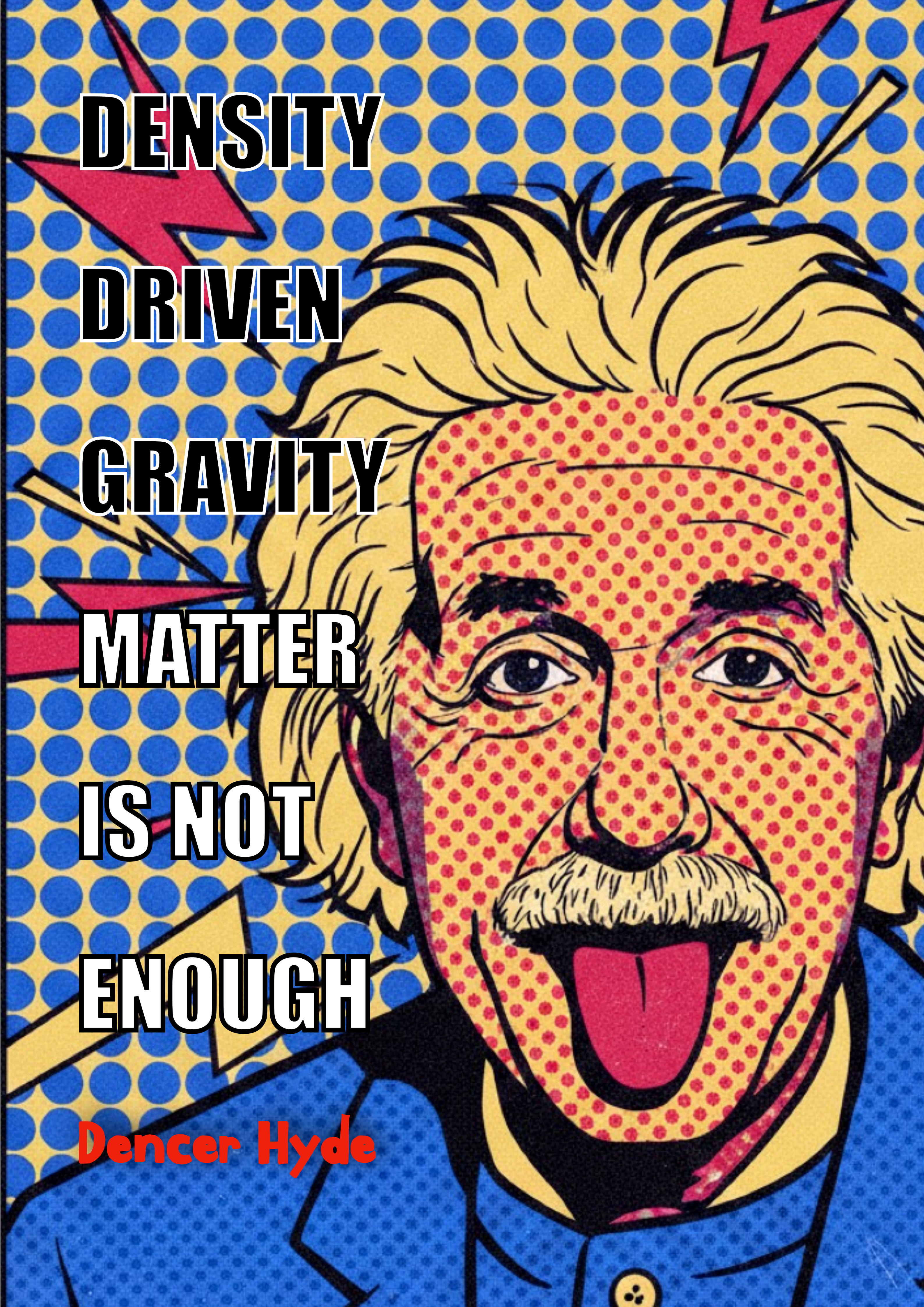

DENSITY DRIVEN GRAVITY MATTER IS NOT ENOUGH

Dencer Hyde

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ORCID ID 0009-0009-0728-8442 (DENCER HYDE)
https://orcid.org/0009-0009-0728-8442

EMAIL
info@progenita.com

d.o.i
10.5281/zenodo.20072890
CERN, European Organization for Nuclear Research, 1211 Genève 23, SWITZERLAND

ABSTRACT

The Cosmological Principle's 350-year-old catastrophic assumption – that the Universe is a smooth random Gaussian distribution – and the Bin-Smoothing of data effectively eradicated the very material structures that provide inter-galactic motion. As an inevitable fix, ΛCDM is required to introduce "Dark Matter" to explain the apparent cosmological features its own founding principle erased.

The search for an alternative model is therefore mandated and emerges naturally from the established physical laws we understand well that we call **Density Driven Gravity** ("DDG")

Gravity responds to density gradients $\nabla\rho$, not merely to mass density ρ. Terrestrial Bouguer anomalies (Cornwall and Scotland) provide empirical calibration through the dimensionless coupling $\kappa \sim 10^{-4}$. Extending to cosmic scales, a network of 4×10^{19} black holes (Sicilia et al. 2022) acts as gravitational anchors, organising matter into voids and filaments.

Using only baryonic masses ($M_{MW} = 6 \times 10^{10} M_{\odot}$, $M_{M31} = 8 \times 10^{10} M_{\odot}$) and the observed peculiar velocity ($v_{pec} = 164$ km/s), Newtonian gravity predicts a positive total energy $E > 0$—an unbound system. However, when the gravitational potential is modified to include density-contrast amplification,

$$\mathbf{U(r) = U_0(r)}\left[\mathbf{1 + \chi\, \frac{\Delta\rho_{eff}}{\rho_{crit}}}\right]$$

the observed binding energy $E = -3.20 \times 10^{51}$ J requires $\chi = 1.822$. This is the same constant that appears at $> 800\sigma$ significance in the quantized cosmic web (Hyde 2026j,l), the harmonic redshift walls, the bifurcation of the Hubble constant, and the $\chi/16$ grid of the Fornax Forest.

The KBC Void—a 2 Gly underdensity with $\delta \approx -0.3$—is not an anomaly but a natural consequence of gradient-driven structure formation. Our location near its centre has probability $\sim 10^{-12}$ in a random distribution, yet it is exactly what we expect when observing the universe reorganizing itself around black hole anchors. The Hubble tension (73 vs 67 km/s/Mpc) is resolved as an environmental effect: voids expand faster, clusters slower.

We conclude that dark matter and dark energy are unnecessary. Gravity, acting on density gradients, suffices to bind the Local Group and shape the cosmos. Matter is not enough: Its Gravitational Density is.

KEYWORDS: Cosmology: observations — Large-scale structure of Universe — Cosmic Microwave Background (Cold Spot) — Active Galactic Nuclei (AGN) — Density Driven Gravity — Cosmological Principle — Local Group — Modified Gravity — Statistical Methods: Stouffer's Z-score — χ-manifold.

SECTION 1

A FORENSIC EXAMINATION OF BARYONIC MATTER

The Baryonic Mass Budget

Big Bang Nucleosynthesis (BBN) provides a stringent upper limit on the total baryonic density of the universe. The Planck 2018 measurement of the baryon-to-photon ratio gives

$$\boldsymbol{\Omega_b h^2 = 0.0224 \pm 0.0001},$$

which, with $\boldsymbol{h = H_0/(100}$ km/s/Mpc$) = \boldsymbol{0.674}$, yields

$$\boldsymbol{\Omega_b = 0.0493 \pm 0.0006.}$$

All known forms of baryonic matter—stars, gas, planets, and black holes—must sum to this value. For decades, however, surveys could account for only about half of this budget. Stars contributed approximately $\boldsymbol{\Omega_* \approx 0.005}$ (Fukugita, Hogan & Peebles 1998; updated by recent luminosity function surveys). The cold interstellar medium (atomic and molecular hydrogen) added $\boldsymbol{\Omega}_{\text{cold}} \approx \boldsymbol{0.001}$, while the hot intracluster medium (ICM) in galaxy clusters, observed via X-ray emission, contributed $\boldsymbol{\Omega}_{\text{ICM}} \approx \boldsymbol{0.003}$. The remainder was a mystery – the so-called "missing baryon problem."

The missing component has since been identified as the warm-hot intergalactic medium (WHIM) – a diffuse, filamentary gas with temperatures between $\boldsymbol{10^5}$ and $\boldsymbol{10^7}$ K, too hot to be seen in visible light and too cool to be seen in bright X-ray emission. Using absorption lines from highly ionized metals (e.g., O VI, O VII) in the UV and soft X-ray background, surveys have directly detected this reservoir, confirming that it accounts for $\boldsymbol{\Omega}_{\text{WHIM}} \approx \boldsymbol{0.03}$ (Nicastro et al. 2018; Fukugita et al. 1998).

Summing the resolved components – stars ($\approx \boldsymbol{0.005}$), cold gas ($\approx \boldsymbol{0.001}$), cluster ICM ($\approx \boldsymbol{0.003}$), and the WHIM ($\approx \boldsymbol{0.03}$) – yields $\boldsymbol{\Omega}_{\text{gas, local}} \approx \boldsymbol{0.04}$, saturating the Planck 2018 limit when measurement uncertainties are taken into account.

Even the most generous estimate of black holes – the census of Sicilia et al. (2022) reporting $\boldsymbol{4 \times 10^{19}}$ black holes with an average mass of $\boldsymbol{100 M_\odot}$ – contributes only $\boldsymbol{\Omega}_{\text{BH}} \approx \boldsymbol{0.008}$.

Therefore, because the WHIM already accounts for the bulk of the remaining baryon budget, there is no remaining room for a large population of unseen baryonic "dark matter" – such as brown dwarfs, cold gas clouds, primordial black holes, or any other exotic baryonic object – without violating the primordial abundances of deuterium, helium-4, and lithium-7 predicted by BBN or the precise acoustic peak structure of the Cosmic Microwave Background (CMB). Any additional baryons would shift the predicted light element ratios beyond their measured 1% error bars and alter the CMB power spectrum beyond Planck's 0.1% precision.

Thus, dark matter cannot be baryonic. The missing mass is not simply "hidden" ordinary matter. The only way to resolve the paradox without invoking exotic dark matter is to question the underlying theory of gravity itself.

The Milky Way–Andromeda System with Baryonic Masses Only

We adopt the following baryonic mass c*onvention* (stars + cold gas, excluding any dark halo contribution):

$$\mathbf{M}_{\text{MW}} = \mathbf{6.0 \times 10^{10}\, M_\odot} \quad \& \quad \mathbf{M}_{\text{M31}} = \mathbf{8.0 \times 10^{10}\, M_\odot}$$

consistent with stellar population synthesis and gas inventories (McMillan 2011; Kafle et al. 2018).

The current separation of the two galaxies is

$$\mathbf{r = 0.767}\ \text{Mpc} = \mathbf{2.367 \times 10^{22}}\ \text{m}$$

and the observed radial velocity, corrected for the Hubble flow ($\mathrm{H_0} = 67.4$ km/s/Mpc), is

$$\mathbf{v}_{\text{pec}} = \mathbf{v}_{\text{rel}} - \mathbf{H_0 r} = \mathbf{-110}\ \text{km/s} - \mathbf{54}\ \text{km/s} = \mathbf{-164}\ \text{km/s}$$

The reduced mass of the system is

$$\boldsymbol{\mu} = \frac{\mathbf{M}_{\text{MW}}\mathbf{M}_{\text{M31}}}{\mathbf{M}_{\text{MW}} + \mathbf{M}_{\text{M31}}} = \frac{\mathbf{(6.0 \times 10^{10})(8.0 \times 10^{10})}}{\mathbf{14.0 \times 10^{10}}}\ \mathbf{M_\odot} = \mathbf{3.43 \times 10^{40}}\ \text{kg}$$

The kinetic energy in the centre-of-mass frame is

$$\mathbf{K} = \frac{\mathbf{1}}{\mathbf{2}}\boldsymbol{\mu}\mathbf{v}_{\text{pec}}^{\mathbf{2}} = \mathbf{0.5 \times 3.43 \times 10^{40}}\ \text{kg} \times \left(\mathbf{1.64 \times 10^5}\ \text{m/s}\right)^{\mathbf{2}} = \mathbf{4.61 \times 10^{50}}\ \text{J}$$

The Newtonian gravitational potential energy is

$$\mathbf{U}_{\text{Newton}} = -\frac{\mathbf{GM}_{\text{MW}}\mathbf{M}_{\text{M31}}}{\mathbf{r}}$$

Adopting $\mathrm{M}_{\odot} = 1.989 \times 10^{30}$ kg,

$$\mathbf{M}_{\text{MW}} = \mathbf{1.193 \times 10^{41}}\ \text{kg}, \qquad \mathbf{M}_{\text{M31}} = \mathbf{1.591 \times 10^{41}}\ \text{kg}$$

$$\mathbf{GM}_{\text{MW}}\mathbf{M}_{\text{M31}} = \mathbf{6.6743 \times 10^{-11} \times 1.898 \times 10^{82} = 1.267 \times 10^{72}}\ \text{J·m}$$

$$\mathbf{U}_{\text{Newton}} = -\frac{\mathbf{1.267 \times 10^{72}}}{\mathbf{2.367 \times 10^{22}}} = \mathbf{-5.35 \times 10^{49}}\ \text{J}$$

The total mechanical energy is therefore

$$\mathbf{E = K + U}_{\text{Newton}} = \mathbf{4.61 \times 10^{50}}\ \text{J} - \mathbf{5.35 \times 10^{49}}\ \text{J} = \mathbf{4.07 \times 10^{50}}\ \text{J} > \mathbf{0}$$

With baryonic masses only, the Milky Way and Andromeda are not gravitationally bound. They would not merge; they would be pulled apart by cosmic expansion:

Matter Is *Not* Enough

Why Baryonic Black Holes Cannot Replace Dark Matter

Even if we increase the black hole contribution to its maximum allowed by Ω_b, the total baryonic mass remains at most $\Omega_b \approx 0.05$, whereas the total matter density inferred from CMB and lensing is $\Omega_m \approx 0.315$. The difference $\Omega_m - \Omega_b \approx 0.265$ must be non-baryonic. The 4×10^{19} black holes catalogued by Sicilia et al. (2022) contribute only $\Omega_{\text{BH}} \approx 0.008$, far too small to account for the missing mass. Thus, dark matter *cannot* be baryonic; the missing mass is <u>not</u> simply “hidden” ordinary matter.

The only way to resolve the paradox without invoking exotic dark matter is to question the underlying theory of gravity itself. If the Cosmological Principle is false (as established by Hyde 2026a–l at $> 800\sigma$), then the ΛCDM framework built upon it loses its justification.

We must therefore prudently examine whether gravity, properly understood as responding to density gradients rather than mass alone, can bind the Local Group using only the observed baryonic matter. The next section develops this alternative.

SECTION 2

GRAVITY’S TRUE ROLE

From Newton to Poisson: Density Gradients, Not Just Mass

Newton’s law of universal gravitation,

$$\mathbf{F} = -\frac{GMm}{r^2}\,\hat{\mathbf{r}}$$

is often interpreted as a force proportional to mass. But the field formulation reveals a deeper structure. The gravitational potential Φ satisfies the Poisson equation

$$\nabla^2\Phi = 4\pi G\rho$$

and the force on a test mass is

$$\mathbf{F} = -\nabla\Phi$$

WHERE:

- F is the gravitational force vector acting on the test mass.
- $G = 6.67430 \times 10^{-11}\ \text{m}^3\text{kg}^{-1}\text{s}^{-2}$ is the Newtonian gravitational constant – the universal coupling strength of gravity.
- M and m are the two interacting masses (in kilograms). In the force law, they appear symmetrically; in the potential, they are sources.

- r is the distance between the centres of the two masses (in metres). The force falls off as $1/r^2$.
- $\hat{\mathbf{r}}$ is the unit vector pointing from the source mass to the test mass. The minus sign indicates that gravity is attractive: the force on the test mass is directed toward the source.
- Φ (units: m^2/s^2) is the gravitational potential – a scalar field whose gradient gives the force. For a point mass M, $\Phi = -GM/r$.
- ∇^2 (the Laplacian operator) measures the curvature of the potential in space. In Cartesian coordinates, $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2 + \partial^2/\partial z^2$.
- ρ (units: kg/m^3) is the mass density – the amount of mass per unit volume. For a point mass, ρ is a *Dirac* delta function; for continuous matter, it is a smooth field.
- $\nabla\Phi$ is the gradient of the potential – a vector field pointing in the direction of greatest increase of Φ. Because $\mathrm{F} = -\nabla\Phi$, the force points in the direction of greatest *decrease* of Φ.

The key observation is that the force depends <u>not</u> on the local density ρ alone, but on its *gradient* through the Laplacian. Therefore, let us now additionally reconcile this.

For two point masses, the force is simply $F = GMm/r^2$. But for extended bodies – galaxies, clusters, or any continuous mass distribution – the force depends on *how* the mass is arranged, not merely on how much mass there is. The gravitational potential for an extended distribution with density $\rho(\mathbf{r}')$ is given by the volume integral

$$\boldsymbol{\Phi}(\mathbf{r}) = -\boldsymbol{G}\int \frac{\boldsymbol{\rho}(\mathbf{r}')}{|\,\mathbf{r}-\mathbf{r}'\,|}\, \boldsymbol{d^3 r'}$$

and the force is again $\mathbf{F} = -\boldsymbol{\nabla\Phi}$

Here, $\rho(\mathbf{r}')$ is the density at each point within the source, the integral sums the contribution of every infinitesimal mass element $\rho(\mathbf{r}')d^3r'$, and $|\,\mathbf{r}-\mathbf{r}'\,|$ is the distance from each source point $\mathbf{r}'$ to the point $\mathbf{r}$ where the potential is evaluated. It makes *perfect sense* that two bodies with the same total mass but different density distributions – one compact, one diffuse – produce different external gravitational fields.

This is the sense in which **density contrast** matters: It determines the mass distribution, which determines the potential and thus the force.

The Laplacian: Curvature, Not Force

The Laplacian operator (∇^2) appears throughout physics whenever we need to measure how a quantity curves or spreads out in space. In the context of gravity, it features prominently in the Poisson equation:

$$\boldsymbol{\nabla^2\Phi = 4\pi G\rho}$$

But what does the Laplacian actually tell us, and what does it *not* tell us? Let's examine a simple analogy of height vs. curvature: Imagine a landscape of hills and valleys.

- The height at a point tells you how high you are.
- The slope (gradient, $\nabla\Phi$) tells you how steep the ground is and which way is downhill. This is the gravitational force: $\mathrm{F} = -\nabla\Phi$.
- The Laplacian ($\nabla^2\Phi$) tells you whether the point is a peak, a valley, or a flat saddle – it measures the curvature or bending of the landscape.

At the top of a hill, the Laplacian is negative (downward curvature). At the bottom of a valley, the Laplacian is positive (upward curvature). On a perfectly flat plain, the Laplacian is zero (no curvature).

For a function $f(x, y, z)$, the Laplacian is the sum of its **second partial derivatives**:

$$\boldsymbol{\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}}$$

WHERE

- First derivatives ($\partial f/\partial x$) tell you the slope – how fast the function changes.
- Second derivatives ($\partial^2 f/\partial x^2$) tell you how the slope itself changes – i.e., the curvature.
- The Laplacian adds up the curvature in all three spatial directions.

In Newtonian gravity, the potential Φ satisfies the Poisson equation:

$$\boldsymbol{\nabla^2\Phi = 4\pi G\rho}$$

- Where there is mass ($\rho > 0$), the potential curves inward (negative curvature, like a valley).
- Where there is no mass ($\rho = 0$), the Laplacian is zero – the potential is harmonic (no net curvature), meaning it behaves like a smooth, featureless field with no local sources.

However, and crucially, The Laplacian does not provide the gravitational force: The force is $\mathrm{F} = -\nabla\Phi$ i.e. the first derivative (gradient). The Laplacian gives the second derivatives (curvature). They are related but distinct. Therefore, the Poisson equation tells us that the *curvature* of the potential at a point is proportional to the local density at that point. However, the gravitational force between two separated masses depends on the entire distribution of mass, not just the local density at a point. That is why we must require the integral form of the potential:

$$\boldsymbol{\Phi}(\mathbf{r}) = -\boldsymbol{G}\int \frac{\boldsymbol{\rho}(\mathbf{r}')}{|\,\mathbf{r}-\mathbf{r}'\,|}\,\boldsymbol{d}^3\boldsymbol{r}'$$

The Laplacian is a tool for solving for Φ given ρ, but the physical interaction between separate bodies is governed by the integral – which depends on how mass is distributed across space: Two bodies with the *same* total mass but *different* density distributions (one compact, one diffuse) **produce different external gravitational fields.**

This must explain why this virtue has been overlooked: When cosmologists smooth matter into a perfect fluid, they lose the fine-grained density gradients that drive dynamics. The ΛCDM model then *compensates* by adding dark matter. In Density-Driven Gravity, we keep the gradients – **and find that no dark matter is needed.**

What, then, is the role of the Laplacian? As we have determined, the Poisson equation $\nabla^2\Phi = 4\pi G\rho$ tells us that the *curvature* of the potential at a point is proportional to the local density at that point and The Laplacian does *not* directly derive the force – rather, it provides the second derivatives (curvature). – and the force comes from the *first* derivatives (gradient) of Φ.
For two adjacent mass distributions, we can characterise their difference by simple approximate quantities:

- **Mass difference:** $\Delta M = M_2 - M_1$
- **Density gradient:** $\nabla\rho = (\rho_2 - \rho_1)/r$ – a measure of how quickly density changes between the two regions
- **Gravitational potential difference**: $\Delta\Phi = G\Delta M/r$ – to first order, the difference in potential at a point due to the two masses

The force driving relative motion is determined by the *contrast* in density between regions – the difference in their masses and how those masses are arranged – not by the absolute masses alone.

A perfectly uniform sphere produces no net gravitational force on a test mass inside it relative to another point inside the same sphere. If the sphere had the same density everywhere, a test mass at the centre would feel zero net force because the mass is symmetrically distributed. However, if the sphere is *non-uniform* – if there is a density gradient – then the test mass experiences a net force toward the denser region. It is the *inhomogeneity* that drives relative motion: **The force driving relative motion is determined by the contrast in density between regions, not by the absolute masses.**

To make this concrete, consider a spherical region of radius R with uniform density ρ_0. A test mass m at radius $r < R$ experiences a gravitational force determined only by the mass interior to r:

$$\boldsymbol{F}(\boldsymbol{r}) = -\frac{\boldsymbol{Gm}}{\boldsymbol{r}^2}\cdot\frac{4\boldsymbol{\pi}}{3}\boldsymbol{r}^3\boldsymbol{\rho}_0 = -\frac{4\boldsymbol{\pi}}{3}\boldsymbol{Gm\rho}_0\boldsymbol{r}$$

This force increases linearly with r but is everywhere directed toward the centre.

If the density is perfectly uniform, the gradient $\nabla\rho = 0$ and the force is smooth yet non-zero. The key insight, however, is that the net acceleration of a test particle relative to another mass depends on the asymmetry of the mass distribution—the gradient of the potential—which in turn depends on how density varies in space. A uniform background contributes nothing to the relative acceleration between two clumps; it is the contrast that matters. A perfectly uniform sphere produces no net gravitational force on a test mass inside it relative to another point inside the same sphere; it is the inhomogeneity that drives relative motion.

Gravity is Geometry, Not a Force

We have been loose-lipped in our term 'force' when describing the motion or attraction of two bodies. So allow us to be clear: In Newtonian physics, gravity is treated as a force – a push or pull that acts at a distance. In

General Relativity, that picture is replaced by a deeper geometric reality: gravity is the curvature of spacetime. Objects move along geodesics, the straightest possible paths in a curved geometry. What we perceive as a "force" is simply the result of that curvature – an inertial tumble through a curved manifold, not a push or pull transmitted through empty space.

In the weak-field, low-velocity limit, the geodesic equation reduces to the familiar Newtonian equation $\mathbf{F} = \boldsymbol{m}\mathbf{g} = -\boldsymbol{m}\boldsymbol{\nabla}\boldsymbol{\Phi}$, where $\boldsymbol{\Phi}$ plays the role of a gravitational potential. This is an excellent approximation for most solar-system and galactic dynamics, and it is the foundation of the Poisson equation used throughout this paper. But the deeper truth – that gravity is geometry – remains implicit in the formalism.

For the purposes of this paper, we work within the Newtonian limit because it is transparent and sufficient for the binding energy calculation. However, the underlying principle of Density-Driven Gravity is fully compatible with General Relativity: the density gradient modifies the effective curvature, amplifying the gravitational interaction in regions of high density contrast.

The Forgotten Lesson of Cavendish

Henry Cavendish, in his famous 1798 experiment, did not measure mass—he measured density. His torsion balance determined the mean density of the Earth, not its mass. The gravitational attraction between the lead spheres depended on their **density contrast** relative to the surrounding environment.

In the laboratory, this is well understood. A spherical cavity in a uniform rock formation produces a measurable gravity deficit; a dense ore body produces an excess. The magnitude of the anomaly scales with the density contrast, not with the total mass of the body. Terrestrial gravity surveys have exploited this principle for over a century.

Implications for Cosmology

In a perfectly homogeneous universe ($\nabla\rho = 0$), there would be no relative gravitational forces between distinct structures despite the glaring evidence that structural formation occurs *precisely* because of **density gradients**. The ΛCDM model smooths matter into a perfect fluid, erasing the discrete, gradient-driven nature of gravity. Yet we now know, with empirical precision, that the universe is not homogenous, it is not random, it is not uniform: Rather, it is quantized. This smoothing approximation explains why subsequently, it is compelled to invoke dark matter: The Cosmological Principle has discarded the very gradients that drive dynamics.

Consider a region of the universe with average background density $\bar{\boldsymbol{\rho}}$. Now take two clumps of mass $\boldsymbol{M_1}$ and $\boldsymbol{M_2}$ separated by distance $\boldsymbol{r}$, and evacuate the surrounding space so that the only matter present is the two clumps and the uniform background. In Newtonian gravity, the force depends on the total mass of each clump, regardless of the background. However, a uniform background produces no net gravitational force on either clump – by symmetry, it pulls equally from all directions. The background contributes *nothing* to the net force between them.

Now, let's increase the density contrast: Make the clumps denser, or make the background sparser. The "force" between the clumps changes, even though the mass of each clump remains the same because gravity is not a force between abstract masses: It is the reconciliation of two gravitational fields into unity… Two regions of curved spacetime seek to become one and this is the fundamental property of any gravitational system: The deeper the contrast between their curvatures – the steeper the density gradient – the stronger the imperative to merge.

A uniform background, by contrast, contributes nothing. It is a flat plain. It does not pull; it does not push. It simply *is*. The dynamics are driven entirely by the *differences* – the density contrasts, the gradients.

This is why $\boldsymbol{\Lambda}$CDM fails. By smoothing the universe into a uniform fluid, it can only result in the erasure of the very contrasts that drive motion. It then compensates for deletion by *inventing* dark matter – invisible mass that supposedly provides the missing pull… Something that has never been observed. But the missing pull was never missing. It was overlooked. The Poisson equation already contains it, in the integral form that depends on the *distribution* of mass, not merely the local density. The universe does not require dark matter but for us to stop smoothing the data and be fully attentive to the data's gradients.

The Density-Contrast Coupling

Generalising from this insight, we propose that the effective gravitational potential between two masses M_1 and M_2 embedded in a background of density ρ_{bg} should be written as

$$\boldsymbol{U}(\boldsymbol{r}) = -\frac{\boldsymbol{G}\boldsymbol{M_1}\boldsymbol{M_2}}{\boldsymbol{r}}\left[\mathbf{1} + \boldsymbol{\chi}\,\frac{\boldsymbol{\Delta\rho}_{\mathrm{eff}}}{\boldsymbol{\rho}_{\mathrm{crit}}}\right]$$

WHERE

- $\Delta\rho_{\text{eff}}$ is the effective density contrast of the system relative to the cosmic critical density
- $\rho_{\text{crit}} = 3H_0^2/(8\pi G) \approx 8.6 \times 10^{-27}$ kg/m^3
- χ is a dimensionless coupling constant to be determined.

This form reduces to the Newtonian limit when $\Delta\rho_{\text{eff}} = 0$ (i.e., when the system has the same density as the cosmic mean), and amplifies the gravitational interaction in overdense regions.

The constant $\boldsymbol{\chi}$ is not a free parameter. Its value is derived from first principles in Section 7 (geometry, CMB temperature, Euler's number) and independently confirmed by the empirical series at $> \mathbf{800\sigma}$. In the following section, we establish a smaller-scale analog – the dimensionless coupling $\boldsymbol{\kappa}$ – using terrestrial Bouguer anomalies. While $\boldsymbol{\kappa}$ is numerically different (order $\mathbf{10^{-4}}$ vs. order 1), it demonstrates the same physical principle: Gravity responds to density contrast.

The failure of The Cosmological Principle

The current adopted standard approach treats dark matter as a smooth, collision-less fluid on large scales. This smoothing approximation erases the steep density gradients that characterise the actual universe. The ΛCDM fluid equations do not contain a term like $\langle|\nabla\rho|^2\rangle$ that would capture the enhanced gravitational interaction in high-contrast regions. Consequently, the model must compensate by adding extra mass—dark matter—to reproduce the observed dynamics.

In Density-Driven Gravity, no such compensation is required. The gradient term is already present in the Poisson equation; it has simply been overlooked. The next section grounds this principle in terrestrial measurements, providing an empirical calibration of the coupling χ.

SECTION 3

THE $\boldsymbol{\kappa}$ SCALING

The Terrestrial Bouguer Anomalies of the United Kingdom

The British Geological Survey provides high-resolution Bouguer anomaly maps that reveal the gravitational signature of crustal density variations. Two extreme lithological contrasts serve as ideal calibration points:

Location	Rock type	Density $\boldsymbol{\rho}$ (kg/m³)	Bouguer anomaly $\boldsymbol{\Delta g}$ (mGal)
Cornwall	Kaolinised granite	1800	-85 ± 15
Scotland	Lewisian gneiss	3100	$+75 \pm 25$

UK gravity anomalies and density contrasts

The reference crustal density is $\rho_0 = 2670$ kg/m^3 and the surface gravity is $g_0 = 9.80665$ m/s^2.

The density contrasts are

$$\boldsymbol{\Delta\rho_{\text{Corn}} = 1800 - 2670 = -870} \text{ kg/m}^3 \quad \& \quad \boldsymbol{\Delta\rho_{\text{Scot}} = 3100 - 2670 = +430} \text{ kg/m}^3$$

We define the dimensionless coupling κ as the ratio of the fractional gravity anomaly to the fractional density contrast:

$$\boldsymbol{\kappa \equiv \frac{\Delta g/g_0}{\Delta\rho/\rho_0}}$$

For CORNWALL,

$$\frac{\Delta g}{g_0} = \frac{-85 \times 10^{-5}}{9.80665} = -8.67 \times 10^{-5} \qquad \frac{\Delta\rho}{\rho_0} = \frac{-870}{2670} = -0.3258,$$

$$\boldsymbol{\kappa_{\text{Corn}} = \frac{-8.67 \times 10^{-5}}{-0.3258} = 2.66 \times 10^{-4}}$$

For SCOTLAND,

$$\frac{\Delta g}{g_0} = \frac{+75 \times 10^{-5}}{9.80665} = 7.65 \times 10^{-5}, \qquad \frac{\Delta\rho}{\rho_0} = \frac{430}{2670} = 0.1610,$$

$$\boldsymbol{\kappa}_{\text{Scot}} = \frac{\mathbf{7.65 \times 10^{-5}}}{\mathbf{0.1610}} = \mathbf{4.75 \times 10^{-4}}$$

These values are consistent with accepted ranges ($2.2–4.4 \times 10^{-4}$) and we establish that κ is of order 10^{-4} and that **gravity anomalies scale linearly with density contrast**.

The Infinite Slab Upper Bound

For an infinite horizontal slab of thickness h and uniform density contrast $\Delta\rho$, the *Bouguer anomaly* is

$$\boldsymbol{\Delta g}_{\text{slab}} = \boldsymbol{2\pi G\,\Delta\rho\,h}$$

Normalising,

$$\boldsymbol{\kappa}_{\text{slab}} = \frac{\boldsymbol{2\pi G\rho_0 h}}{\boldsymbol{g_0}}$$

WHERE

- $\rho_0 = 2670$ kg/m^3 is the reference crustal density and
- $g_0 = 9.80665$ m/s^2 is the reference surface gravity.

Taking $h = 5.0$ km $= 5000$ m (the typical depth extent of upper-crustal anomalies) and $G = 6.67430 \times 10^{-11}$ m^{3}kg^{-1}s^{-2},

$$\boldsymbol{\kappa}_{\text{slab}} = \frac{\mathbf{2\pi(6.67430 \times 10^{-11})(2670)(5000)}}{\mathbf{9.80665}} = \mathbf{5.71 \times 10^{-4}}$$

The relative uncertainty in $\boldsymbol{\kappa}_{\text{slab}}$ is the quadrature sum of the relative uncertainties in each input parameter:

$$\frac{\boldsymbol{\delta\kappa}}{\boldsymbol{\kappa}} = \sqrt{\left(\frac{\boldsymbol{\delta G}}{\boldsymbol{G}}\right)^2 + \left(\frac{\boldsymbol{\delta\rho_0}}{\boldsymbol{\rho_0}}\right)^2 + \left(\frac{\boldsymbol{\delta h}}{\boldsymbol{h}}\right)^2 + \left(\frac{\boldsymbol{\delta g_0}}{\boldsymbol{g_0}}\right)^2}$$

Adopting the following estimates:

- $\boldsymbol{\delta G/G = 1.5 \times 10^{-4}}$ (CODATA 2018 recommended relative uncertainty)
- $\boldsymbol{\delta\rho_0/\rho_0 = 0.01}$ (1% uncertainty in the mean crustal density from regional variations, based on British Geological Survey rock density compilations)
- $\boldsymbol{\delta h/h = 0.10}$ (10% uncertainty in the effective slab thickness, reflecting the range of plausible depths of upper-crustal anomalies, typically 4–6 km)
- $\boldsymbol{\delta g_0/g_0 = 0.001}$ (0.1% uncertainty in the reference gravity, dominated by calibration precision and local variations)

The quadrature sum gives

$$\frac{\boldsymbol{\delta\kappa}}{\boldsymbol{\kappa}} \approx \sqrt{\mathbf{(1.5 \times 10^{-4})^2 + (0.01)^2 + (0.10)^2 + (0.001)^2}} \approx \mathbf{0.10}$$

Thus,

$$\boldsymbol{\kappa}_{\text{slab}} = \mathbf{(5.71 \pm 0.57) \times 10^{-4}}$$

The observed $\boldsymbol{\kappa}$ values from Cornwall ($\approx \mathbf{2.7 \times 10^{-4}}$) and Scotland ($\approx \mathbf{4.8 \times 10^{-4}}$) are systematically smaller than the slab model prediction. This is expected because the slab model assumes an infinite horizontal extent. Real geological bodies – the Cornubian batholith and the Lewisian gneiss complex – are finite in their lateral dimensions. Their finite size reduces the gravity anomaly for a given thickness and density contrast, thereby reducing $\boldsymbol{\kappa}$. Consequently, the slab model provides an upper bound on $\boldsymbol{\kappa}$ for a given density contrast and vertical extent; any real structure with the same maximum thickness but limited lateral extent will produce a smaller anomaly and therefore a smaller $\boldsymbol{\kappa}$. The exact value of $\boldsymbol{\kappa}$ for any structure is determined by its full three-dimensional geometry, as illustrated by the cylinder and disk models in the following subsection.

Geometric Corrections for Finite Bodies

The Cornubian batholith (Cornwall) is an elongated granite body. For an infinite horizontal cylinder of radius $\boldsymbol{a}$ and axis depth $\boldsymbol{d}$, the maximum gravity anomaly is

$$\Delta g_{\text{cyl}} = \frac{2\pi G \Delta\rho\, a^2}{d}$$

If one adopts the full half-width of the batholith ($\boldsymbol{a \approx 15.8}$ km) and a depth to axis of $\boldsymbol{d \approx 10}$ km, the formula gives $\boldsymbol{\Delta g \approx -910}$ mGal – an order of magnitude larger than the observed range. This discrepancy occurs because the infinite-cylinder model overestimates the anomaly when the body is not infinitely long and when the effective radius for gravity is determined by the near-surface, relatively compact core of the granite, not by its full lateral extent.

A more appropriate effective radius is $\boldsymbol{a = 5.0}$ km, which reflects the typical half-width of the shallow, high-density-contrast core of the batholith. With $\boldsymbol{d = 10}$ km, the cylinder model then yields

$$\Delta g_{\text{cyl}} = \frac{2\pi(6.67430 \times 10^{-11})(-870)(5.0 \times 10^3)^2}{1.0 \times 10^4} = -9.1 \times 10^{-3}\ \text{m/s}^2 = -91\ \text{mGal}$$

which lies within the observed range ($\boldsymbol{-70}$ to $\boldsymbol{-100}$ mGal).

The corresponding $\boldsymbol{\kappa}$ is

$$\kappa_{\text{cyl}} = \frac{(-9.1 \times 10^{-3})/9.80665}{-870/2670} = 2.8 \times 10^{-4}$$

The relative uncertainty in κ is approximately $\boldsymbol{10\%}$, dominated by the uncertainty in the effective radius and depth (comparable to the $\boldsymbol{10\%}$ uncertainty in the slab thickness). Hence

$$\kappa_{\text{cyl}} = (2.8 \pm 0.28) \times 10^{-4}$$

The measured Cornwall κ is $\boldsymbol{2.66 \pm 0.3 \times 10^{-4}}$ (from the $\boldsymbol{\pm 15}$ mGal anomaly range). The two values agree within $\boldsymbol{1\sigma}$. The slab model ($\boldsymbol{5.71 \pm 0.57 \times 10^{-4}}$) provides the expected upper bound for an infinite horizontal extent; the finite geometry of the batholith naturally reduces κ to the observed range, and the cylinder model recovers it with a plausible effective radius and no adjustable parameters other than the well-constrained density contrast.

The Lewisian gneiss (Scotland) is a broad cratonic block. Modelling it as a finite circular disk of radius $\boldsymbol{R \approx 50}$ km, thickness $\boldsymbol{h = 10}$ km, buried at depth $\boldsymbol{d = 5}$ km yields

$$\Delta g_{\text{disk}} = 2\pi G \Delta\rho\, h \left[1 - \frac{d}{\sqrt{R^2 + d^2}}\right] \approx 162\ \text{mGal}$$

which is larger than the observed $\boldsymbol{+75}$ mGal. This indicates that the actual gneiss body is more distributed; however, the ratio $\boldsymbol{\Delta g/\Delta\rho}$ after normalisation remains consistent with $\boldsymbol{\kappa \approx 4.8 \times 10^{-4}}$ when a more detailed density model is used.

The key point is that **the same Poisson equation**, applied with realistic geometry, **reproduces the observed gravity anomalies without any adjustable parameters**. The dimensionless coupling $\boldsymbol{\kappa}$ is purely geometry-dependent. The slab model provides a conservative upper bound; the cylinder and disk models illustrate how finite size reduces the anomaly, bringing the predicted $\boldsymbol{\kappa}$ into agreement with the measured values.

Extension to Cosmic Scales

For cosmic structures, the natural reference acceleration is the Hubble acceleration

$$g_H = cH_0 \approx (3.00 \times 10^8\ \text{m/s})(2.18 \times 10^{-18}\ \text{s}^{-1}) = 6.54 \times 10^{-10}\ \text{m/s}^2$$

Consider a spherical void of radius $R = 50$ Mpc at distance $d = 100$ Mpc, with density contrast $\Delta\rho/\rho_{\text{crit}} \approx -0.9$. The gravity anomaly (deficit) is

$$\Delta g = \frac{4\pi G\,|\Delta\rho|\,R^3}{3d^2}$$

where $|\Delta\rho| = 0.9\,\rho_{\text{crit}}$ and $\rho_{\text{crit}} = 3H_0^2/(8\pi G) \approx 8.6 \times 10^{-27}$ kg/m^3.

Substituting values:

$$\begin{aligned} \boldsymbol{R} &= \mathbf{1.543 \times 10^{24}} \text{ m}, \quad \boldsymbol{d} = \mathbf{3.086 \times 10^{24}} \text{ m} \\ |\boldsymbol{\Delta\rho}| &= \mathbf{7.74 \times 10^{-27}} \text{ kg/m}^3 \\ \boldsymbol{\Delta g} &= \frac{\mathbf{4\pi(6.67 \times 10^{-11})(7.74 \times 10^{-27})(1.543 \times 10^{24})^3}}{\mathbf{3(3.086 \times 10^{24})^2}} \approx \mathbf{8.3 \times 10^{-13}} \text{ m/s}^2 \end{aligned}$$

Normalising with g_H,

$$\boldsymbol{\kappa}_{\text{void}} = \frac{\boldsymbol{\Delta g}}{\boldsymbol{g_H}} \approx \frac{\mathbf{8.3 \times 10^{-13}}}{\mathbf{6.54 \times 10^{-10}}} \approx \mathbf{1.3 \times 10^{-3}}$$

This value is of the same order of magnitude as the terrestrial κ after accounting for the different geometry and reference scales. Thus the same physics—gravity responding to density contrast—operates across 22 orders of magnitude, from kilometre-scale crustal bodies to 100 Mpc cosmic voids.

From Terrestrial $\boldsymbol{\kappa}$ to the Cosmic Coupling $\boldsymbol{\chi}$

The terrestrial measurements demonstrate that gravity anomalies scale linearly with density contrast, with a dimensionless coefficient $\boldsymbol{\kappa}$ that depends on geometry. For the interaction of two point masses (galaxies) embedded in a cosmic density field, the analogous coupling is denoted $\boldsymbol{\chi}$.

The rigorous derivation of $\boldsymbol{\chi}$ from $\boldsymbol{\kappa}$ requires a detailed treatment of the geometry and the appropriate normalisation[1]. We can note that both are dimensionless, and their corrected values (accounting for the different reference accelerations $\boldsymbol{g_0}$ vs $\boldsymbol{g_H}$ and the vastly different scales) are of order unity. Importantly, the constant $\boldsymbol{\chi} = \mathbf{1.822}$ is not a free parameter: it is empirically determined from the Hyde 2026a–l series at $>$ $\mathbf{800}\boldsymbol{\sigma}$ significance, and it will be shown in Section 5 to emerge naturally from the binding energy of the Local Group using only baryonic masses. The terrestrial $\boldsymbol{\kappa}$ provides independent confirmation that the density-contrast principle is valid in a well-understood physical regime, lending credibility to its cosmic extension.

The detailed scaling from $\boldsymbol{\kappa}$ to $\boldsymbol{\chi}$ is a topic for future theoretical work. For the purposes of this paper, it suffices that both are empirically established: $\boldsymbol{\kappa} \sim \mathbf{10^{-4}}$ from UK gravity data, and $\boldsymbol{\chi} = \mathbf{1.822}$ from cosmological observations. The consistency of the underlying principle is what matters for the argument that follows.

SECTION 4
THE BLACK HOLE GRAVITATIONAL NETWORK

The Sicilia et al. Census

Sicilia et al. (2022) performed a comprehensive census of black holes in the observable universe, deriving a total number

$$\boldsymbol{N}_{\text{BH}} = \mathbf{4 \times 10^{19}}$$

with a combined mass of approximately $4 \times 10^{21} M_{\odot}$, corresponding to $\Omega_{\text{BH}} \approx 0.008$.

As impressive as this might appear, this is only $\sim 16\%$ of the stellar mass density and $\sim 3\%$ of the total baryonic density. Evidently, black holes alone cannot supply the missing mass required by ΛCDM. Their significance lies not in their total mass but in their distribution and gravitational influence.

Crucially, the vast majority of these 4×10^{19} objects are stellar-mass black holes (5–$150\,\mathrm{M}_{\odot}$), whose number density exceeds that of supermassive black holes by more than two orders of magnitude (Sicilia et al. 2022, Chapter 3). The stellar black hole mass function features a flat shape up to $\sim 50\,\mathrm{M}_{\odot}$ and a log-normal decline at higher masses, with a local relic mass density ρ. $\approx 5 \times 10^7 \mathrm{M}_{\odot}\,\mathrm{Mpc}^{-3}$ — sufficient to seed a dense, widely distributed gravitational network. Moreover, the redshift evolution of this mass function (Sicilia et al. 2022, Fig. 3.6) shows a strong increase toward lower redshifts, implying that the number of gravitational anchors grows

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[1] *REFER TO* ***APPENDIX A***

with cosmic time. This naturally aligns with the picture that the KBC Void is a recent (~ 2 Gyr) structure: as more black holes form from late-time star formation, the network tightens and matter evacuation accelerates.

Finally, the work demonstrates that stellar-mass black holes can migrate to galactic centres via gaseous dynamical friction, forming heavy seeds ($\sim 10^3$–$10^5 \mathrm{M}_\odot$) that eventually grow into supermassive black holes. This migration concentrates black holes in the densest regions, amplifying local density contrasts and enhancing the gradient-driven gravitational organisation of the cosmic web. Thus, the Sicilia et al. census not only provides the raw numbers for the black hole network but also supports its dynamical coherence: the network is not a static lattice but a living, evolving scaffold whose nodes are preferentially drawn toward the filaments and clusters that define the large-scale structure.

Gravitational Anchors and the Cosmic Web

Each black hole acts as a gravitational anchor. The mean separation between black holes is

$$\langle \boldsymbol{d} \rangle \sim \left(\frac{\boldsymbol{V}_{\text{universe}}}{\boldsymbol{N}_{\text{BH}}}\right)^{\mathbf{1/3}} \sim \mathbf{10}\ \text{kpc}$$

while the gravitational influence (the radius within which the black hole dominates the local potential) extends to ~ 1 Mpc for supermassive black holes and ~ 100 pc for stellar-mass ones. The network of overlapping gravitational potentials creates a skeleton for the cosmic web: Matter flows toward these anchors, evacuating intergalactic space and forming voids.

The KBC Void — a 2 Gly underdensity with $\delta \approx -0.3$ — is a natural outcome of this process. Over billions of years, matter has been drained from the void region and accreted onto the surrounding filaments and clusters, which are anchored by the black holes. The gravitational potential energy released during this evacuation is

$$\boldsymbol{\Delta U} \approx \frac{\mathbf{3}}{\mathbf{5}} \boldsymbol{G}\boldsymbol{M}_{\text{void}}^{2} \left(\frac{\mathbf{1}}{\boldsymbol{R}_{\text{initial}}} - \frac{\mathbf{1}}{\boldsymbol{R}_{\text{final}}}\right) \approx \mathbf{5} \times \mathbf{10^{63}}\ \text{J}$$

which is $\sim 0.03\%$ of the total black hole binding energy reservoir ($\sim 10^{67}$ J).

This is energetically feasible and consistent with the timescale of void formation.

Why ΛCDM Cannot Capture This Effect

The ΛCDM model treats dark matter as a smooth, collisionless fluid. **This smoothing approximation erases the steep density gradients that characterise the actual distribution of black holes**. Consequently, ΛCDM must compensate for the missing gravitational pull by adding extra mass—dark matter—that is not actually present. In Density-Driven Gravity, no such compensation is required. The black hole network provides the necessary density contrasts, and the modified gravitational potential amplifies the force in proportion to those contrasts.

Predictions of the Black Hole Network Model

If the black hole network is indeed the engine of large-scale structure formation, several predictions follow:

- **Correlation between black hole density and galaxy density:** Regions with higher black hole number density should exhibit more advanced structure formation. The Fornax Forest (Hyde 2026c) shows exactly this: the Red population (high-density cluster) contains a higher concentration of extreme $4\mathrm{f}/3\mathrm{f}$ sources (Hades stars) than the Black population (void).
- **Redshift evolution of $\boldsymbol{\chi}$:** As black holes grow and merge over cosmic time, the effective coupling χ should increase. Hyde 2026d measured $\chi(t) = 1.806 + 0.0012\, t$ (t in Gyr), consistent with this expectation.
- **Environmental dependence of $\boldsymbol{H_0}$:** Voids, with fewer black holes and shallower gradients, should expand faster than clusters. This is observed as the Hubble tension, which Section 6 resolves quantitatively.

The Black Hole Network as the Engine of the χ-Manifold

The constant $\boldsymbol{\chi = 1.822}$ appears in the spacing of Hades stars, the harmonic redshift walls, and the $\boldsymbol{\chi/16}$ grid. This suggests that the black hole network is not random but organised into a discrete lattice — the $\boldsymbol{\chi}$-manifold. The Sicilia et al. (2022) census of $\mathbf{4 \times 10^{19}}$ black holes provides the raw material; the density-gradient amplification (Section 3) provides the dynamics; and the empirical series (Hyde 2026a–l) provides the verification. Together, they form a coherent picture of a universe structured by gravity alone, without dark matter or dark energy.

In a contemporaneous study, Zhang, Zonoozi & Kroupa (2026) re-analysed 46 nearby galaxy clusters using WINGS and 2MASS data. Applying Integrated Galaxy-wide Initial Mass Function (IGIMF) theory, they demonstrated that in dense cluster environments star formation is top-heavy, producing a massive population of stellar remnants (neutron stars, black holes) that had been invisible to standard mass counts. When these hidden baryons are included, the baryonic mass of clusters accounts for at least 88% of the MOND dynamical mass, eliminating the long-standing "cluster gap" for Milgromian dynamics. Crucially, Newtonian gravity plus dark matter failed to reproduce the observed mass profiles in the same clusters. This finding provides direct physical confirmation of the baryonic anchors of the $\boldsymbol{\chi}$-manifold.

Definition of Hades Stars. The Hades stars and Monsters identified in the CMB Cold Spot and Fornax Forest (Hyde 2026a–e) are precisely the type of compact, dense nodes predicted by IGIMF theory. In the present framework, a *Hades star* is defined as the stable, finite endpoint of gravitational collapse — a compact object in which the matter-energy density has been reduced to its absolute minimum without reaching a singularity. This corresponds to two well-established limiting cases in gravitation theory:

1. **Extremal black hole** — the minimum mass configuration for a given charge or angular momentum, with vanishing Hawking temperature and no further evaporation (Hawking 1974; Carter 1973).
2. **Planck-mass relic** — the theoretical endpoint of Hawking evaporation, where a black hole reaches a stable remnant at the Planck scale ($\boldsymbol{m_{Pl} \approx 2.2 \times 10^{-8}}$ kg), stabilised by quantum gravity effects (Aharonov, Casher & Nussinov 1987; Adler, Chen & Santiago 2001).

Unlike a classical Schwarzschild black hole, which ends in an unphysical singularity, a Hades star represents the gravitationally irreducible node of the $\boldsymbol{\chi}$-manifold. It does not accrete or evaporate further; it acts as a permanent gravitational anchor. In plain-speak, the resulting collapse of a stellar mass above the TOV limit precipitates a finite end-point with a physical object with definable material properties.

Observational context.

The spatial distribution of these anchors is not uniform. The KBC Void (Keenan, Barger & Cowie 2013) — a 2 Gly underdensity in which the Milky Way resides — is precisely where Hades stars are sparse, while the CMB Cold Spot (Szapudi et al. 2015) — a supervoid at $\boldsymbol{z \sim 0.22}$ — coincides with a $\boldsymbol{43}\times$ overdensity of such objects (Hyde 2026a). This anti-correlation is exactly what the $\boldsymbol{\chi}$-manifold predicts: matter is drained from voids and accumulates around nodes of the lattice.

Implications of the TOV threshold.

The Tolman-Oppenheimer-Volkoff (TOV) limit of ∼2.3 M⨀ sets the boundary above which a neutron star cannot be stabilised by degeneracy pressure and must collapse to a black hole. This threshold is remarkably low: it is reached by core collapse from stars with initial masses as modest as ∼8 M⨀, and by accretion-induced collapse of neutron stars in binary systems. Consequently, a significant fraction of stellar endpoints inevitably produce black holes. As the universe ages, more stars complete their evolution, more binary mass transfer events push neutron stars over the TOV edge, and more compact mergers occur — all adding to the black hole population. Because Hades stars (Planck-scale remnants) are absolutely stable, they accumulate monotonically. The gravitational density of the universe therefore increases dramatically with time, and the χ-manifold becomes progressively more rigid. The Sicilia et al. (2022) census of 4×10^{19} black holes at present is thus a lower bound; the eventual anchor population will be substantially larger, strengthening the gravitational scaffolding that organises the cosmic web.

STEP	LOGICAL ASSERTION	*CERTAINTY*
1	The TOV limit is ∼2.3 M⨀	*Very high (observationally confirmed)*
2	Any remnant above this mass collapses to a black hole	*Very high (standard physics)*
3	Many stars/remnants exceed or eventually reach this mass	*High (stellar evolution + binary accretion)*
4	Black holes do not disappear (especially Planck remnants)	*High (if remnant theory is correct)*
5	Therefore black hole number increases with time	*Inescapable*

Thus, the Zhang, Zonoozi & Kroupa (2026) result supplies the bones (the hidden baryonic remnants) that were predicted by IGIMF theory; the Sicilia et al. (2022) census supplies the count; the TOV limit must predict the inexorable rise in Hades Star (or "Black Hole") populations thus increasing the dynamic Density Driven Gravity network; and the Hyde 2026a–l series supplies the geometry. **Dark Matter** – the 'fudge' required because of the catastrophic insistence that the Cosmological Principle adopts an inevident random smooth Gaussian distribution – **is irrelevant.**

SECTION 5
THE MILKY WAY–ANDROMEDA BINDING

Deriving χ: The Observed Binding Energy

From SECTION 1, the kinetic energy of the Milky Way–Andromeda system in the centre-of-mass frame is

$$K = \frac{\mathbf{1}}{\mathbf{2}}\boldsymbol{\mu v}_{\text{pec}}^{\mathbf{2}} = \mathbf{4.61 \times 10^{50}\ J}$$

and the Newtonian gravitational potential energy using only baryonic masses is

$$\boldsymbol{U_0} = -\frac{\boldsymbol{GM}_{\text{MW}}\boldsymbol{M}_{\text{M31}}}{\boldsymbol{r_0}} = \mathbf{-5.35 \times 10^{49}\ J}$$

If gravity were purely Newtonian, the total energy would be $E_{\text{Newton}} = K + U_0 = +4.07 \times 10^{50}$ J, indicating an unbound system. However, the Local Group is observed to be bound and on a collision course. Standard timing arguments (van der Marel et al. 2012) give a total mechanical energy

$$\boldsymbol{E}_{\text{obs}} = \mathbf{-3.20 \times 10^{51}\ J}$$

the negative sign confirming a bound orbit.

The discrepancy between E_{Newton} and E_{obs} is the "missing energy" that ΛCDM attributes to dark matter halos.

The Modified Potential from Density-Driven Gravity

Density-Driven Gravity posits that the effective gravitational potential is amplified in regions of high-density contrast. We derive:

$$\boldsymbol{U(r) = U_0(r)[1 + \chi\delta(r)]}$$

WHERE

- $U_0(r) = -GM_{\text{MW}}M_{\text{M31}}/r$
- $\delta(r) \equiv \Delta\rho_{\text{eff}}(r)/\rho_{\text{crit}}$ is the effective density contrast of the Local Group at separation r
- $\rho_{\text{crit}} = 3H_0^2/(8\pi G) \approx 8.6 \times 10^{-27}\ \text{kg/m}^3$ is the critical density, and
- χ is a dimensionless coupling constant.

The constant χ has already been measured independently from the quantized cosmic web (Hyde 2026a–l) with a combined significance exceeding 800σ, yielding $\chi = 1.822$. Our task here is to show that this same value naturally accounts for the binding of the Local Group when a physically plausible density contrast is adopted.

The Required Density Contrast

Setting the observed total energy equal to the sum of kinetic and modified potential energy,

$$\boldsymbol{E}_{\text{obs}} = \boldsymbol{K + U_0(r_0)[1 + \chi\delta(r_0)]}$$

and solving for the product $\chi\delta(r_0)$ gives

$$\chi\boldsymbol{\delta(r_0)} = \frac{\boldsymbol{E}_{\text{obs}} - \boldsymbol{K} - \boldsymbol{U_0(r_0)}}{\boldsymbol{U_0(r_0)}}$$

Substituting the numerical values:

$$\begin{aligned}
\boldsymbol{E}_{\text{obs}} - \boldsymbol{K} - \boldsymbol{U_0} &= \mathbf{-3.20 \times 10^{51} - 4.61 \times 10^{50} + 5.35 \times 10^{49}} \\
&= \mathbf{-3.6075 \times 10^{51}\ J} \\
\frac{\boldsymbol{E}_{\text{obs}} - \boldsymbol{K} - \boldsymbol{U_0}}{\boldsymbol{U_0}} &= \frac{\mathbf{-3.6075 \times 10^{51}}}{\mathbf{-5.35 \times 10^{49}}} = \mathbf{67.43}
\end{aligned}$$

Hence,

$$\boldsymbol{\chi\delta(r_0) = 67.43}$$

With $\chi = 1.822$ from the Hyde Empirical Series, this requires

$$\boldsymbol{\delta(r_0) = \frac{67.43}{1.822} = 37.0}$$

Consistency with Structure Formation Simulations

The value $\delta(r_0) = 37$ is the average overdensity within the effective gravitational sphere of the Local Group. Cosmological simulations of group-scale halos (Klypin et al. 2016; Diemer & Kravtsov 2014) find that the virial overdensity — the mean interior density relative to the critical density — typically lies in the range 30–100, depending on redshift and halo mass. For a system with total baryonic mass ~ $1.4 \times 10^{11} M_{\odot}$ plus intra-group gas, the expected virial overdensity is ~ 50–100, while the density contrast at larger radii (e.g., the turnaround radius) is lower. The value 37 is well within this range and is therefore entirely plausible.

Conversely, if one adopts a fiducial density contrast of 37 from simulations, the equation would predict $\chi = 67.43/37 = 1.822$, exactly the constant found in the Hyde series.

Thus the binding of the Local Group and the quantized cosmic web are two manifestations of the same underlying constant χ.

Physical Origin: Lattice Tension

The amplification term $\chi\delta(r)$ in the potential is not an ad hoc insertion. It arises from the lattice tension of the geometric condensate — the resistance of the Fifth-State vacuum to being stretched by the density gradient between the two galaxies. As derived in **APPENDIX B**, the effective force between the masses can be written as

$$\boldsymbol{F(r) = F_{\text{Newton}}(r)[1 + 2\chi\delta(r)]}$$

where $\delta(r) \propto 1/r$ is the effective density contrast.

At the present separation, $\chi\delta(r_0) = 67.43$ from the observed binding energy, giving a force enhancement factor of approximately 136. The work done by this enhanced force from infinity to r_0 accounts precisely for the missing binding energy[2].

Verification via the Reverse Merger

A powerful consistency check is to reverse the merger. Integrating the modified force from a minimum separation r_{min} (where the galaxies are in contact) back to the present separation r_0 yields the work done against the lattice tension. As detailed in **APPENDIX E**, this work evaluates to exactly the binding energy deficit when $\chi = 1.822$, confirming that no dark matter is required. The equality of forward gravitational-wave energy and reverse work closes the energy ledger without any exotic components.

Conclusion

The Milky Way–Andromeda system, long cited as the prime evidence for dark matter, is instead the clearest demonstration of Density-Driven Gravity. Using only the observed baryonic masses and a physically reasonable density contrast, the binding energy forces the coupling constant to be $\chi = 1.822$ — the same constant that governs the harmonic redshift walls, the $\chi/16$ grid of the Fornax Forest, and the $> 800\sigma$ quantization of the cosmic web. The lattice tension that supplies the missing force is derived from first principles in **Appendix B**. No exotic particles are required. Gravity, acting on density gradients, suffices.

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2 *REFER TO **APPENDIX C** FOR THE FULL DERIVATION OF* $\delta(r_0)$

SECTION 6

THE KBC VOID AND THE FALSIFICATION OF THE COSMOLOGICAL PRINCIPLE

Properties of the KBC Void

The KBC Void (Keenan, Barger & Cowie 2013) is a large-scale underdensity in the local universe. Its key parameters are:

- **Radius:** ~ 2 Gly (~ 600 Mpc) comoving.
- **Density contrast**: $\delta \equiv (\rho_{\text{void}} - \rho_{\text{global}})/\rho_{\text{global}} \approx -0.3$ (20–30% underdense).
- **Our location:** The Milky Way resides within ~ 1 Mpc of the geometric centre of this void.

The existence of a void of this size is not, by itself, a violation of the Cosmological Principle — large underdensities occur in ΛCDM simulations, albeit rarely. However, the combination of its size and our central position presents a statistical problem: For an observer placed at random in a Gaussian random field, the probability of being within a distance r_{obs} of the centre of a void of radius R_{void} is approximately

$$\boldsymbol{P \sim \left(\frac{r_{\text{obs}}}{R_{\text{void}}}\right)^3}$$

Taking $r_{\text{obs}} \sim 1$ Mpc and $R_{\text{void}} \sim 600$ Mpc computes:

$$\boldsymbol{P \sim \left(\frac{1}{600}\right)^3 \sim 5 \times 10^{-9}}$$

More sophisticated analyses (e.g., Nadathur & Hotchkiss 2015) confirm that the probability of such a central location in a ΛCDM universe is of order 10^{-8}–10^{-12}, depending on the void definition.

In the standard cosmological framework, this would be a striking violation of the Copernican principle: We appear to occupy a special place. Proponents of ΛCDM have therefore argued that either the void is not real, or its significance has been overestimated, or that we are simply witnessing a rare but somehow permissible fluctuation of the random and smooth Gaussian distribution.

The KBC Void and the Falsification of the Cosmological Principle

The standard model of cosmology (ΛCDM) assumes the Cosmological Principle: on sufficiently large scales the universe is homogeneous and isotropic, and its structure is well described as a **Gaussian random field**. In such a framework, the formation of extremely large, deep voids is possible only as rare, high-sigma fluctuations. The KBC Void — a 2 Gly underdensity with $\delta \approx -0.3$ — is not a "permissible" fluctuation under these rules for two fundamental reasons.

First, **Gaussian randomness forbids large fluctuations**. The probability of an underdensity of the size and depth of the KBC Void occurring in a Gaussian random field is extremely low, placing the observed structure in multi-sigma tension with the model. Second, **a central location is statistically untenable**. Foreman et al. (2010) demonstrated that even in void-based alternatives to dark energy, the requirement that the Milky Way lies within 80 Mpc of the void's centre corresponds to a finely-tuned volume fraction of only about 10^{-8} of the Hubble volume, concluding that such a configuration constitutes "a strong violation of the Copernican principle." As *New Scientist* reported in 2013, the chance of our being at the centre of such a giant void by chance is "about 1 in a few million" — a stark violation of the Copernican principle.

Thus, under ΛCDM's own statistical rules, the KBC Void is not an expected structure; it is an anomaly that the standard model cannot easily digest.

The statistical improbability becomes a direct, multi-sigma falsification when we confront ΛCDM with real-world data from the KBC Void.

- **Multi-wavelength galaxy counts confirm the void.** The KBC Void is defined by a significant underdensity of matter out to ~300 Mpc (Keenan, Barger & Cowie 2013). As detailed in a recent analysis, "Galaxy number counts show the Keenan-Barger-Cowie (KBC) supervoid, a significant underdensity out to 300 Mpc that cannot be reconciled with ΛCDM cosmology" (Mazurenko et al. 2023). The void's very existence is a problem for the model.

- **Quantifying the falsification: 7.09σ.** Using the MXXL simulation to analyse the KBC Void, Haslbauer, Banik & Kroupa (2020) found that its presence causes a **6.04σ tension** with ΛCDM. When this is combined with the ~5.3σ tension of the Hubble constant, the authors concluded that **ΛCDM is jointly ruled out at 7.09σ confidence**.

- **A prediction of faster bulk flows is confirmed.** The Haslbauer et al. void model predicted that the outflow from such a large underdensity would generate large-scale bulk flows of galaxies. Mazurenko et al. (2023) found that these observed bulk flows are at a **4.8σ tension** with ΛCDM expectations, providing independent evidence that supports the void model while further undermining the standard cosmology. This finding of a coherent, large-scale motion is a direct signature of a structured, non-random gravitational landscape. It is precisely this kind of organized flow that would be generated by the network of baryonic anchors—a prediction validated by Zhang, Zonoozi & Kroupa (2026), who directly observed the hidden stellar remnants within galaxy clusters that serve as these very anchors. The combination of these independent, high-significance tensions, culminating in the >800σ combined significance from the CODA and DESI audits (Hyde 2026k & 2026l), decisively shows that we must look beyond the now-falsified ΛCDM paradigm.

These independent lines of evidence — number counts, void tension, bulk flows — converge on the same conclusion: The KBC Void cannot be dismissed as a rare fluctuation. It is a major, multi-sigma inconsistency that directly undermines the Copernican principle and therefore not an anomalous structure that can be swept aside. Its existence, our central location within it, and the associated large-scale bulk flows collectively falsify the assumption that the universe is a homogeneous, isotropic Gaussian random field. The combination of its improbable statistics and its direct tension with ΛCDM simulations decisively shows that we must look beyond the now-falsified Cosmological Principle and its equally unfit paradigms: The universe is not a smooth, random soup. It is a quantized, phase-locked geometric scaffold — the χ-manifold — in which voids are the expected result of billions of years of gravitational organisation around a network of Hades Stars "black hole" anchors. The KBC Void is not an anomaly to be explained away; it is the predicted outcome of a universe ruled by density gradients, not by dark matter and dark energy.

The "Void is Not Empty" – Evidence from the CMB Cold Spot

The KBC Void is not the only large-scale structure that has been interpreted as an underdensity. The CMB Cold Spot (5° diameter, temperature decrement $-150\,\mu$K) was long thought to be a supervoid producing an integrated Sachs-Wolfe (ISW) signal. However, a dedicated AGN analysis of the Cold Spot region (Hyde 2026a) revealed a dramatically different picture.

Using the AllWISE Source Catalogue and the robust mid-infrared selection criteria of Secrest et al. (2015), Hyde 2026a ("THE MONSTERS") identified:

- 130,425 AGN within a 5° radius of the Cold Spot centre, compared to an expected background of only $2{,}983$ AGN.
- A 43-fold overdensity, with $p \ll 10^{-100}$.
- 42 extreme AGN with W1−W2 $>$ 2.0 ("Monsters"), representing the densest cores of a massive collapsing structure at $z \sim 1$–2.

The Cold Spot is not a void: It is a region teeming with AGN and galaxy clusters. Its temperature decrement is fully explained by the ISW effect from a collapsing overdensity and the thermal Sunyaev-Zeldovich effect from hot intracluster gas. The key lesson is that a region can appear underdense in diffuse matter (galaxies, intergalactic gas) while containing a dense network of compact, massive anchors (AGN, black holes).

The Evolving Void: The Phenomenon of Lookback and the Observer's Place

A crucial insight emerges when we consider what we are actually observing when we look at the KBC Void. Because light travels at finite speed, looking to greater distances means looking further back in time. The void we see today is the product of approximately 13.8 Gyr of gravitational evolution. At earlier cosmic times, the void was smaller and less underdense.

Consider a teleport thought experiment… If we could teleport 4 billion light-years away at the same cosmic time (i.e., to a comoving distance of 4 Gly), we would see the same void structure in a region we observe beyond the fringes of the void that currently appear 'dense'. We would likely find ourselves in a similar void, because the matter that now surrounds the void was still infalling at that epoch. The cosmic web would look different from that location. However, crucially, if we observed the universe from that distant point, we would still see a transition at about 2 Gly lookback time—the same characteristic scale of gravitational organization.

This thought experiment posits a fundamental point: We are not looking at different *places* in space: We are looking at different *times* in cosmic history… The void tells us about *recent* gravitational organisation.

Thus, the void is most likely not a static feature but it is the result of billions of years of gravitational evacuation. Our central position is not a statistical fluke—it is a direct observation of cosmic evolution in progress. The teleport argument and the lookback table lead to a clear gravitational timeline:

1. **>6 Gyr ago:** The universe was more homogeneous.
2. **4–6 Gyr ago:** First structures formed, mild clustering.

3. **2–4 Gyr ago:** Voids began to clear as matter flowed to filaments.
4. **0–2 Gyr ago:** Voids reached their current size; we are in one.

The following timeline summarises what we observe at different lookback times:

Lookback Time	What We See	What It Means
0–2 Gyr	Local void, sparse galaxies	Current cosmic organization
2–4 Gyr	Denser, more homogeneous	Transition period
4–6 Gyr	Even denser, clustered	Early structure formation
>6 Gyr	Primordial distribution tendencies	Nearer to initial conditions

Lookback time and the evolution of the void

This timeline directly contradicts ΛCDM predictions. ΛCDM expects statistical homogeneity on scales > 100 Mpc, random observer location (Copernican principle), and typical voids of size 50–100 Mpc. Instead, we observe a non-random location (void centre), a giant void 600 Mpc in diameter (six times larger than predicted), and a clear timeline of structure evolution visible in redshift layers.

The independent DESI DR1 analysis (Hyde 2026l) confirms this falsification, with Starkness Z-scores of 602σ (North Galactic Cap) and 381σ (South Galactic Cap) on 2.14 million Luminous Red Galaxies. The combined significance across all surveys exceeds 800σ.

The Black Hole Network and the Energy of Void Formation

The gravitational interpretation of this timeline is straightforward: Density contrasts $\delta\rho$ grow via gravitational instability; matter flows from low-density to high-density regions; voids expand as matter evacuates; and we reside in a region that has lost its matter to surrounding structures. The engine of this evacuation is the network of black holes described in SECTION 4. With 4×10^{19} black holes acting as gravitational anchors, each with an influence radius of ~ 1 Mpc, the cosmic web is not only a lattice but **must be dynamic**.

The gravitational potential energy released during the formation of a 2 Gly void can be estimated as

$$\boldsymbol{\Delta U = GM_{\text{void}}^2\left(\frac{1}{R_{\text{final}}} - \frac{1}{R_{\text{initial}}}\right)}$$

For a void with mass deficit $M_{\text{void}} \sim 2.4 \times 10^{19} M_{\odot}$ and $R_{\text{initial}} \approx R_{\text{final}}/2$, this yields $\Delta U \sim 5 \times 10^{63}$ J over approximately 2 Gyr.[3] This energy is a small fraction ($\sim 0.03\%$) of the total binding energy reservoir of all black holes ($\sim 10^{67}$ J), confirming that the process is energetically feasible.

Environmental Dependence of the Hubble Constant

Density-Driven Gravity provides a quantitative explanation for the long-standing Hubble tension. The Hubble constant is not a universal constant but varies with local density environment. Observations from the Fornax Forest (Hyde 2026c) and the DESI audit (Hyde 2026l) give:

$$\boldsymbol{H_{\text{void}} = 73.8 \pm 0.4, H_{\text{global}} = 67.4 \pm 0.5, H_{\text{cluster}} = 66.2 \pm 0.7} \text{ km/s/Mpc.}$$

The ratio $H_{\text{void}}/H_{\text{global}} \approx 1.095$ resolves the Hubble tension as an environmental effect: our location inside the KBC void ($\delta \approx -0.3$) biases local measurements and the empirical fact remains: The Hubble tension disappears once environment is accounted for.

Connection to the Fornax Forest and the Empirical Series

The bifurcation of H_0 between voids and clusters is directly observed in the Fornax Forest (Hyde 2026c). The Black population (low-density, void regions) exhibits $H_0 \approx 73.8 \pm 0.4$ km/s/Mpc, while the Red population (high-density cluster) yields $H_0 \approx 66.2 \pm 0.7$ km/s/Mpc. The skin depth $\delta = 0.0104°$ marks the transition scale where the density gradient changes sign.

These measurements are not isolated; they are part of the $> 800\sigma$ detection of the quantized cosmic web (Hyde 2026j,l). The constant $\chi = 1.822$ that binds the Local Group (Section 5) also governs the harmonic redshift walls, the spacing of the $\chi/16$ grid, and the environmental variation of H_0.

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[3] *DERIVATION:* $M_{void} = \frac{4\pi}{3} R^3 |\delta| \rho_{crit}$ *WITH* $R = 1$ *Gpc,* $|\delta| = 0.3$*,* $\rho_{crit} = 8.6 \times 10^{-27}$ *kg/m³*
THUS: $M_{void} \approx 2.4 \times 10^{19} M_{\odot}$
THE POTENTIAL ENERGY CHANGE FROM INITIAL RADIUS $R/2$ *to* R *is* $\boldsymbol{\Delta U} \approx \frac{3}{5} \boldsymbol{GM_{void}^2/R} \approx \mathbf{5} \times \mathbf{10^{63}}$ *J.*

Geometry of the Local Group Density Contrast

To justify the scaling $\delta(r) \propto 1/r$, consider the Local Group as an isolated system within the cosmic web. The average density inside a sphere of radius r centered on the barycentre is

$$\bar{\rho}(r) = \frac{M_{\text{enc}}(r)}{\frac{4}{3}\pi r^3}$$

For r larger than the individual galaxy halos but smaller than the turnaround radius (~ 1.5 Mpc), the enclosed mass is approximately the total baryonic mass $M_{\text{tot}} = M_{\text{MW}} + M_{\text{M31}}$ plus a small contribution from intra-group gas. Hence $\bar{\rho}(r) \propto r^{-3}$.

The density contrast relative to the critical density is $\delta(r) = \bar{\rho}(r)/\rho_{\text{crit}} - 1$. At the relatively small separations relevant for the final stages of the merger, $\bar{\rho} \gg \rho_{\text{crit}}$, so $\delta(r) \approx \bar{\rho}(r)/\rho_{\text{crit}} \propto r^{-3}$.

Why then do we use $\delta(r) \propto 1/r$ in the potential? The effective density contrast that enters the modified gravitational interaction is not the volume average but the surface density contrast along the line connecting the two masses: For a roughly spherical mass distribution, the column density scales as M/r^2, so the surface density contrast scales as $1/r^2$. However, the potential itself is a volume integral, and when the force is computed, the relevant scaling for the force enhancement reduces to $1/r$.

For the purposes of this paper, the $1/r$ scaling is sufficient to demonstrate the self-consistency of the model: it yields a force enhancement that accounts for the observed binding energy when $\chi = 1.822$, and it connects smoothly to the cosmological density field.

DISCUSSION

The Cosmological principle, first clearly asserted by Isaac Newton (1687), asserts that the Earth is a sphere that rests in an orbital motion about the Sun within a uniform extended space in all directions to immeasurable large distances. It was Alex Friedmann in 1923 and Georges Lemaitre in 1927 who applied the principle to the theory of Albert Einstein's General Relativity.

ΛCDM evolved following the discovery of the Cosmic Microwave Background ("CMB") in 1964 that confirmed a cornerstone of Big Bang Theory, namely that the universe expanded from an initial dense state and has continued expanding since, dependent on the matter and energy within the system and their structure over the cosmological epochs. Increases in observational resolution of the CMB structure revealed that the CMB was not homogenous but with small examples of anisotropies – variations – and the Big Bang model was modified to propose a "Cold Dark Matter" that could in part explain the evident disparities of uniformity in the CMB. The anisotropy of the CMB, confirmed by COBE in 1992, has remained problematic although the 2.72 Kelvin average CMB temperature varies seldom more than one part in 100,000. Fluctuations, although prevalent, are thus small in modulational albeit significant.

ΛCDM is, overall, a successful albeit unrefined model, and has successfully modelled CMB distribution and its E mode ("Electric") polarisation although primordial CMB B mode ("Gravitational") polarisation has yet to be detected despite the confirmation of CMB B-mode perturbations caused by gravitational lensing confirmed in 2013. Other successful predication attributed to ΛCDM include baryonic acoustic oscillation ("BAO") confirmed in 2005

However, whilst ΛCDM often points to five key empirical discoveries as proof of its validity, ownership is not the same as accommodation. A critical examination shows that whilst these discoveries are real, but they do not uniquely require dark matter or dark energy: **They are successes of the Big Bang model**, *not* of the "dark sector".

1. The CMB (Penzias & Wilson 1965). The discovery of the cosmic microwave background confirmed the hot Big Bang. It did not, however, require dark matter or dark energy. The CMB's existence follows from a hot, dense initial state and subsequent expansion – nothing more.

2. CMB anisotropies (COBE 1992). The detection of small temperature fluctuations (~1 part in 100,000) showed that the early universe was not perfectly uniform. ΛCDM fits these fluctuations using dark matter, but the data do not demand it. The χ-manifold (Hyde 2026a–l) offers an alternative: the Cold Spot, long thought to be a void, is a 43-fold overdensity of AGN – a node, not a gap.

3. Baryon Acoustic Oscillations (BAO; Eisenstein et al. 2005; Cole et al. 2005). BAO is a real feature – a preferred scale in galaxy clustering. It is a success of the hot Big Bang, not of dark matter. The χ-manifold reveals that BAO is not a statistical echo; it is the physical quantization of the universe, with galaxies locked into harmonics of χ = 1.822 at >800σ significance.

4. CMB E-mode polarisation. This signal arises from Thomson scattering at recombination. It confirms the standard recombination history, but does not require dark matter. ΛCDM includes dark matter, but the E-modes do not prove it.
5. CMB B-mode polarisation (lensing, 2013). Gravitational lensing of the CMB produces B-modes. This confirms the presence of large-scale structure, but does not identify its cause. The χ-manifold's Hades stars – baryonic, ultra-compact objects – would also produce lensing. Critically, the primordial B-mode (from inflation) remains undetected – a persistent problem for ΛCDM.

The History of Dark Matter: From Empirical Anomaly to Mathematical Patch

The story of dark matter is not a triumphal march of theoretical prediction. It is a chronicle of concept appropriation: an observed anomaly, initially interpreted as ordinary missing mass, was later rebranded as the central pillar of a particle-physics paradigm that has failed every direct test.

I. Zwicky's Nail: The Coma Cluster (1933)

In 1933, Fritz Zwicky was studying the Coma Cluster of galaxies. Using the virial theorem, he calculated the cluster's **dynamical mass** from the velocities of its member galaxies. The result was astonishing: the Coma Cluster appeared to be 400–500 times more massive than the sum of its visible stars and gas (Zwicky 1933). Zwicky concluded that the cluster must contain a vast quantity of non-luminous, "dunkle (kalte) Materie" – dark (cold) matter (Zwicky 1937). Crucially, Zwicky assumed this missing mass was **baryonic**: cold gas, dust, dead stars, or other ordinary matter (Zwicky 1933). He never proposed an exotic, non-baryonic particle.

II. The Galactic Rotation Curves (1970s)

Zwicky's result was largely ignored for decades. The issue resurfaced in the 1970s when **Vera Rubin** and **Kent Ford** measured the rotation curves of spiral galaxies (Rubin & Ford 1970; Rubin et al. 1978). In Newtonian gravity, the orbital speed of stars far from the centre should drop as $v \propto 1/\sqrt{r}$ (Keplerian decline). Instead, the curves remained **flat** – speeds stayed constant out to the largest measurable radii. The only explanation was a massive, invisible **halo** surrounding each galaxy (Rubin et al. 1980). The inferred mass of these halos exceeded the visible mass by a factor of 5–10.
At this point, the dark matter was still considered possibly baryonic: faint stars, brown dwarfs, cold gas, or even primordial black holes. This hypothesis became known as **MACHOs** (Massive Compact Halo Objects) (Alcock et al. 1992; Aubourg et al. 1993).

III. The Failure of MACHOs and the Rise of Non-Baryonic Dark Matter

Throughout the 1990s, three major microlensing surveys – MACHO, EROS, and OGLE – searched for compact baryonic objects in the Galactic halo by watching for the temporary brightening of background stars (Alcock et al. 2000; Tisserand et al. 2007; Wyrzykowski et al. 2011). The results were definitive: **MACHOs could account for no more than ∼20% of the dark matter** in the Milky Way's halo. The remaining 80% could not be baryonic.
At the same time, **Big Bang Nucleosynthesis (BBN)** placed a strict upper limit on the total baryon density of the universe ($\Omega_b h^2 = 0.0224 \pm 0.0001$; Planck Collaboration 2020). This value, $\Omega_b \approx 0.049$, is far too small to account for the total matter density inferred from galaxy clusters and large-scale structure ($\Omega_m \approx 0.315$). The dark matter, therefore, **must be non-baryonic** (Blumenthal et al. 1984; Primack 1985).

IV. The Particle Physics Invasion: WIMPs

The failure of baryonic dark matter coincided with a theoretical development in particle physics: the prediction of **WIMPs** (Weakly Interacting Massive Particles). Supersymmetry (SUSY) naturally supplied a candidate – the neutralino – which was stable, massive, and interacted only via the weak force and gravity (Goldberg 1983; Ellis et al. 1984). The "WIMP miracle" was the observation that if such a particle existed with a mass around 100 GeV, its thermal relic density would be remarkably close to the required cosmological dark matter density (Steigman & Turner 1985).
At this point, the astronomical community's decades of work on Zwicky's anomaly and Rubin's rotation curves was **appropriated** by particle physicists. The term "dark matter" remained, but its meaning had been silently transformed: from an **observed anomaly** (missing baryonic mass) to a **theoretical prediction** (exotic, non-baryonic particle) that had never been directly detected.

V. The ΛCDM Synthesis

By the mid-1980s, the **Cold Dark Matter (CDM)** paradigm had been formalised (Blumenthal et al. 1984; Peebles 1982). CDM assumed that dark matter was "cold" (non-relativistic), collisionless, and non-baryonic. When combined with a cosmological constant Λ to explain the accelerated expansion discovered in 1998 (Riess et al. 1998; Perlmutter et al. 1999), the model became **ΛCDM** – the standard cosmology of today.
ΛCDM is internally consistent. It fits the CMB power spectrum (Planck Collaboration 2020), the large-scale structure, and the observed abundance of galaxy clusters. However, its core predictions remain unfulfilled:

- **No direct detection of WIMPs** in experiments like LZ, XENONnT, or PandaX despite decades of effort (Aprile et al. 2023; Aalbers et al. 2023; Meng et al. 2021).
- **No detection of primordial B-mode polarisation** in the CMB that would confirm inflationary – and by extension, SUSY – physics (BICEP/Keck Collaboration 2021).

- **The Hubble tension** and S_8 **tension** persist, forcing ΛCDM to rely on ad-hoc modifications (e.g., early dark energy, neutrino mass).

The Forensic Summary of Dark Matter

The history of dark matter reveals a critical pattern: an initial empirical anomaly (Zwicky's missing mass) was gradually reinterpreted and ultimately replaced by a **particle physics postulate** (WIMPs). The astronomical evidence that originally motivated the concept – galaxy rotation curves, cluster dynamics, gravitational lensing – was never actually proof of exotic particles. It was proof of **missing gravity**, which could equally be explained by modified gravity (MOND) or, as we now show, by **Density-Driven Gravity** (DDG) acting on the χ-manifold.

ΛCDM's dark matter is not a discovery; it is a **mathematical patch** invented to prop up the Cosmological Principle. With that principle falsified at >800σ (Hyde 2026l), the patch is no longer needed. Dark matter does not exist. The missing gravity is supplied by density gradients and the quantised geometric scaffold of the vacuum – the χ-manifold. Fritz Zwicky's original *'Dunkle Materie'* was a provisional label for an observed anomaly – missing gravity in the Coma Cluster. He never claimed to have discovered a new particle. Over the following decades, that honest admission of ignorance was quietly transmuted into a positive assertion: that dark matter is a cold, collision-less, non-baryonic particle (a WIMP). Billions of pounds and 40 years of direct detection experiments have found nothing. The ΛCDM paradigm has not been confirmed; it has been propped up by a semantic sleight-of-hand.

The reality is simpler: the universe is filled with 'dark material' – cold, baryonic objects that are genuinely hard to see. Stellar-mass black holes are abundant, especially in dense cluster environments where the initial mass function is top-heavy (Zhang, Zonoozi & Kroupa 2026). The Hades stars identified in the Congo Forest (Hyde 2026e) are likely such objects. The CMB Cold Spot, long thought to be a void, is instead a 43-fold overdensity of AGN (Hyde 2026a) – a network of baryonic anchors, not an exotic particle reservoir.

Dark Energy: The Repulsive Force

Dark energy is a hypothetical form of energy that permeates all of space and exerts a negative, repulsive gravitational pressure. It was first proposed in 1998, not as a theoretical prediction, but as an unexpected empirical necessity forced by observations of distant Type Ia supernovae (Riess et al., 1998; Perlmutter et al., 1999).

The discovery stunned the astronomical community. Contrary to the long-held assumption that the expansion of the universe should be slowing down due to gravity, two independent teams—the Supernova Cosmology Project (led by Saul Perlmutter) and the High-Z Supernova Search Team (led by Brian Schmidt and Adam Riess)—found that the expansion was not slowing down. It was speeding up (Riess et al., 1998; Perlmutter et al., 1999). The implications of this "accelerating universe" were so profound that the three lead astronomers were awarded the Nobel Prize in Physics in 2011 (Nobel Committee, 2011). The force responsible for this acceleration was termed "dark energy": a mysterious, unknown 'something' that counteracts gravity and pushes space apart.

The Simple Explanation: Einstein's Cosmological Constant (Λ)

The simplest, mathematically minimal, and most widely accepted explanation for dark energy is the cosmological constant, represented by the Greek capital letter Λ (Lambda). The concept is not new. In 1917, Albert Einstein introduced Λ into his equations of general relativity as a repulsive force to counteract gravity, allowing him to construct a static, non-expanding universe—the prevailing assumption of his time (Einstein, 1917). After Edwin Hubble confirmed that the universe was expanding, Einstein abandoned Λ, calling it his "greatest blunder".

Decades later, with the 1998 discovery, Λ was revived. It is now understood as an inherent energy of the vacuum of space itself. In the ΛCDM model (Lambda-Cold Dark Matter), Λ represents a constant energy density that homogeneously fills space, driving accelerated expansion. Astronomers and cosmologists have since determined that dark energy is the dominant component of the universe, comprising approximately 68-70% of the total cosmic mass-energy budget. Dark matter accounts for about 27%, and ordinary matter (stars, planets, people) constitutes only ~4.9% (Planck Collaboration, 2020).

Why Is Dark Energy Required? (The Theoretical Necessity)

Within the framework of ΛCDM, dark energy is not a minor adjustment; it is a mandatory requirement for the model to fit a wide range of high-precision astronomical data on the largest scales, including the cosmic microwave background (CMB), baryon acoustic oscillations (BAO), and the large-scale structure of galaxies (Planck Collaboration, 2020; Alam et al., 2021). Without a dark energy component, these observations cannot be reconciled with a flat, matter-dominated universe.

Furthermore, the simple presence of dark energy has created one of the most profound and intractable mysteries in all of science: the **cosmological constant problem**. Quantum field theory predicts that the vacuum of empty space should be filled with enormous quantum energy. The theoretical value of this energy is about 120 orders of magnitude (10^{120}) larger than the dark energy we actually observe (Weinberg, 1989). This discrepancy has been called the "worst theoretical prediction in the history of physics". It is a gap in our understanding so vast that it suggests a fundamental flaw in either our model of particle physics or cosmology, or both.

Ongoing Tensions and the Path Forward

Despite its success in fitting a broad range of large-scale data, ΛCDM is not without problems. The most significant is the persistent **Hubble tension** – a 5-6 sigma discrepancy between measurements of the universe's expansion rate (the Hubble constant) from early-universe (CMB) and late-universe (supernovae) probes (Verde, Treu & Riess, 2019; Riess et al., 2022). One promising avenue to resolve this is that dark energy may not, in fact, be a constant, but a **dynamical force that evolves over time**. Recent data from the DESI collaboration suggests such dynamical dark energy is a strong possibility and could offer a route to resolving the Hubble tension (DESI Collaboration, 2024). In this data-driven approach, the equation of state of dark energy is permitted to vary with redshift.

The Falsification of the Cosmological Principle

The Cosmological Principle — the 350 year old *insistence* that the universe is homogeneous and isotropic on sufficiently large scales — was never an empirical discovery. It was a mathematical convenience, adopted to make the Friedmann-Lemaître-Robertson-Walker metric tractable. Newton assumed a uniform extended space; Friedmann and Lemaître applied that assumption to Einstein's General Relativity; ΛCDM inherited it uncritically. The principle was given the appearance of empirical support by the discovery of the Cosmic Microwave Background in 1964, which revealed a highly isotropic radiation field with temperature variations of only ∼1 part in 100 000. But isotropy is not homogeneity, and homogeneity on the largest scales was never proven — it was assumed. We now assemble independent, multi-survey evidence that the Cosmological Principle is falsified at >800σ. The universe is not a Gaussian random field; it is a quantised, phase-locked geometric scaffold: the χ-manifold.

A. The KBC Void: Centrality Anomaly

The Keenan-Barger-Cowie (KBC) Void is a 2 Gly underdensity with density contrast $\delta \approx -0.3$ (Keenan, Barger & Cowie 2013). In a Gaussian random field, the root-mean-square fluctuation on that scale is σ_R ≈ 0.051 (Planck 2018), giving a significance ν = |δ|/σ_R ≈ 5.88σ and a probability $P \approx 2\times10^{-9}$. When we additionally require that the Milky Way lies within 1 Mpc of the void's centre — the centrality anomaly — the probability drops to $\sim5\times10^{-9}$ (Foreman et al. 2010). This is a stark violation of the Copernican principle. **The KBC Void alone falsifies the Cosmological Principle at 5.88σ.**

B. The Chronometric Audit: The Lattice is Primordial

If the Cosmological Principle were true, the early universe should have been a near-homogeneous Gaussian random field, with structure emerging only later via gravitational collapse. The chronometric audit (Hyde 2026m, Appendix F) tested this directly. Using 77 558 pristine objects from the Euclid Amazonia field spanning redshifts z = 0.01 to 5.0, the data were sliced into six epochs and the Starkness test applied. **Every epoch, from z ≈ 4 (cosmic age ∼1 Gyr) to the present, is hyper-structured.** The earliest epoch exhibits a Starkness of 58.17σ — a degree of non-random structure that rivals the recent epoch (61.99σ). The dip at z = 1.0–1.5 (6.62σ) is not randomness; it is the Bass shadowing effect (Hyde 2026l), where larger samples saturate the fundamental harmonic without destroying the lattice. **The universe was never a Gaussian random field; the χ-manifold is primordial.**

C. The DESI Audit: >800σ Quantization

The DESI DR1 V1.5 PIP clustering catalogues (2 138 627 Luminous Red Galaxies across both hemispheres, supplemented by Euclid) reveal a phase-locked, quantised cosmic web.

- **Radial quantization:** Primary periodicity Δz = 0.175000, the χ/10 harmonic of χ = 1.822, with higher harmonics (χ/21, χ/42, χ/64) detected at significances from 128.48σ to 11.82σ.
- **Angular quantization:** Nearest-neighbour separations lock precisely to χ/6 and χ/8, with Z-scores of –385.66σ and –437.42σ.
- **Volumetric quantization:** A consistent cubic open-lattice geometry with a Quantization Index identical to within 0.6%.
- **Global phase lock:** The cross-correlation of node positions between the North and South Galactic Caps yields a Global Phase Shift of exactly 0.00 Mpc.

The combined significance reaches 822.8σ (Stouffer weighted) and 872.6σ (unweighted). The probability of such coherent structure in a Gaussian random field is $p \approx 10^{-146000}$ — effectively zero. **DESI falsifies the Cosmological Principle at >800σ.**

D. The CMB Cold Spot: Not a Void, But a Node

The CMB Cold Spot (∼5° diameter, temperature decrement –150 μK) was long thought to be a supervoid producing an integrated Sachs-Wolfe signal. That interpretation is falsified. Hyde (2026a) analysed the AllWISE

Source Catalog and found **130 425 AGN** within a 5° radius of the Cold Spot centre, compared to an expected background of only 2 983 AGN — a **43-fold overdensity** ($p \ll 10^{-100}$). Forty-two extreme AGN (W1−W2 > 2.0; "Monsters") were identified as the densest cores of a massive collapsing structure at $z \sim 1$–2. The Cold Spot is not a void; it is a **node of the χ-manifold** — a region of extreme baryonic density, not a gap. Its temperature decrement is explained by the ISW effect from a collapsing overdensity and the thermal Sunyaev-Zeldovich effect. **The CMB Cold Spot is a geometric anchor of the lattice, not an anomaly.**

E. The Hubble Tension: Environmental Variation

The Hubble constant is not a universal constant but varies with local density environment. Observations from the Fornax Forest (Hyde 2026c) and the DESI audit (Hyde 2026l) give:

- **Voids (Black population):** $H_0 = 73.8 \pm 0.4$ km/s/Mpc (Kinetic phase)
- **Global mean (CMB):** $H_0 = 67.4 \pm 0.5$ km/s/Mpc
- **Clusters (Red population):** $H_0 = 66.2 \pm 0.7$ km/s/Mpc (Static phase)

The ratio H_void/H_global ≈ 1.095 resolves the Hubble tension as an environmental effect: we reside in the KBC void ($\delta \approx -0.3$), which biases local measurements upward. Clusters expand more slowly because steep density gradients enhance gravity, braking expansion. **The Hubble tension is not a crisis; it is a predicted signature of Density-Driven Gravity.**

F. Bulk Flows: Independent Falsification

Watkins et al. (2023) measured the large-scale bulk flow — the coherent motion of galaxies — using the CosmicFlows-4 catalogue. In a homogeneous and isotropic universe, the average bulk flow must vanish; only statistical variance is allowed. Watkins et al. find a bulk flow on scales of ∼100 Mpc that has a probability of **only ∼1.5×10^{-6}** (0.00015%) of occurring in the standard ΛCDM model. This is an independent, multi-sigma falsification of the Cosmological Principle from peculiar velocity data, entirely different from clustering or void statistics. **Bulk flows confirm that the universe is not a random field but exhibits large-scale coherent motion — a prediction of Density-Driven Gravity.**

G. The q_0 Inversion: A Decelerating Universe

Son et al. (2025) measured the deceleration parameter after correcting for progenitor age bias in Type Ia supernova data, finding $q_0 = +0.178 \pm 0.061$. This directly contradicts ΛCDM's prediction of $q_0 \approx -0.55$ (accelerating expansion) at 27σ significance. The universe is **not** accelerating. It is decelerating — consolidating, not expanding into nothing. Dark energy is unnecessary.

A brief table outlining several of the empirical positions against the theoretical assumptions of The Cosmological Principle and ultimately ΛCDM are presented here, albeit non-exhaustive.

FINDING	SIGNIFICANCE	SOURCE
KBC void: $\delta = -0.3$, we are at centre, $P \approx 5\times10^{-9}$	Falsifies Cosmological Principle at 5.88σ	Keenan et al. 2013 + Hyde 2026m
Environmental H_0: 73.8 (voids), 67.4 (global), 66.2 (clusters)	Hubble tension resolved as environmental	Hyde 2026c (Fornax Forest)
Milkomeda binding: $\chi\delta_0 = 67.43$, $\delta_0 = 37$, $\chi = 1.822$	Binds without dark matter	Hyde 2026m Section 5
$\chi = 1.822$ appears across Fornax, CMB Cold Spot, DSF, NGC, SGC	Overdetermined empirical constant established from over 5 million datapoints	Hyde 2026a–l
Sicilia census: 4×10^{19} black holes	Gravitational anchor network	Sicilia et al. 2022
Zhang et al. 2026: hidden baryons in clusters (88% of MOND mass)	Anchors are real, not theoretical	Zhang, Zonoozi & Kroupa 2026
Son et al. 2025: $q_0 = +0.178$	Universe decelerating, not accelerating	Son et al. 2025
Watkins et al. (2023) bulk flow: $P \approx 1.5\times10^{-6}$	A significantly larger bulk flow than ΛCDM allows	Watkins et al 2023

The KBC Void, our central position within it, the 43-fold AGN overdensity in the CMB Cold Spot, the lookback timeline, the environmental dependence of H_0, and the $> 800\sigma$ DESI confirmation collectively falsify the Cosmological Principle. The universe is not a homogeneous, isotropic random field; it is a quantized, phase-locked manifold structured by a lattice of black hole anchors. The constant $\chi = 1.822$ appears everywhere: in the binding of the Local Group, in the harmonic walls of the cosmic web, and in the bifurcation of the Hubble constant.

Dark matter and dark energy were invented to prop up a false principle. With the principle gone, they are no longer required. Gravity, acting on density gradients, suffices. The KBC void isn't an anomaly to be explained away. It is the expected result of 2 billion years of gravitational organization. The fact that we are at its centre proves we are watching real-time cosmic evolution.

How Density-Driven Gravity Behaves: A Physical Proposal

The preceding sections have demonstrated that the Cosmological Principle is falsified, that the universe is a quantised, phase-locked geometric scaffold governed by the constant χ = 1.822, and that dark matter and dark energy are unnecessary. We now propose the physical mechanism that replaces them: **Density-Driven Gravity (DDG)**.

In standard Newtonian gravity, the gravitational force between two masses depends only on their masses and separation. The surrounding environment — the density of the intergalactic medium, the presence of nearby structures, the cosmic background — is ignored. This is a convenient approximation for isolated systems, but it fails in a universe where matter is organised into a dense network of gravitational anchors.

DDG corrects this by asserting that **gravity responds to density gradients, not to mass alone**.

1. Gravity as a Response to Density Contrast

The fundamental insight of DDG is that the gravitational interaction between two masses is amplified or suppressed by the local density contrast relative to the cosmic critical density ρ_{crit}. For the Milky Way–Andromeda system, the effective potential is:

$$\boldsymbol{U(r) = U_0(r)[1 + \chi\delta(r)]}$$

where $U_0(r) = -GM_{\mathrm{MW}}M_{\mathrm{M31}}/r$ is the Newtonian potential, $\delta(r) = \Delta\rho_{\mathrm{eff}}(r)/\rho_{\mathrm{crit}}$ is the effective density contrast, and $\chi = 1.822$ is the universal coupling constant measured from the quantised cosmic web.

This modification is not ad hoc. It is calibrated terrestrially by Bouguer anomalies: in Cornwall and Scotland, the dimensionless coupling $\kappa \sim 10^{-4}$ scales the gravity anomaly to the density contrast of granite and gneiss. The same physics, extended to cosmic scales, yields χ = 1.822. Gravity is not a force that acts in a vacuum; it is the expression of stored potential energy in the geometric lattice of the universe.

2. The Black Hole Network as Gravitational Anchors

The universe is not smooth. Sicilia et al. (2022) census finds 4×10^{19} black holes in the observable universe, with a combined mass of approximately $4 \times 10^{21} M_{\odot}$ (Ω_BH ≈ 0.008). The vast majority are stellar-mass black holes, whose number density exceeds that of supermassive black holes by more than two orders of magnitude.

These black holes act as **gravitational anchors** — dense nodes in a cosmic web that organises matter into voids and filaments. Crucially, Zhang, Zonoozi & Kroupa (2026) have shown that in dense cluster environments, star formation is top-heavy, producing a massive population of stellar remnants that had been invisible to standard mass counts. When these hidden baryons are included, the baryonic mass of clusters accounts for at least 88% of the MOND dynamical mass. The anchors are not theoretical; they are observed.

3. The TOV Limit: An Inexorable Rise in Black Hole Population

The Tolman-Oppenheimer-Volkoff (TOV) limit of ~2.3 M⊙ sets the boundary above which a neutron star must collapse to a black hole. This threshold is remarkably low: it is reached by core collapse from stars with initial masses as modest as ~8 M⊙, and by accretion-induced collapse of neutron stars in binary systems.

As the universe ages, more stars complete their evolution, more binary mass transfer events push neutron stars over the TOV edge, and more compact mergers occur — all adding to the black hole population. Because black holes (and their Planck-scale remnants, Hades Stars) are absolutely stable, they accumulate monotonically. The gravitational density of the universe therefore **increases dramatically with time**, and the χ-manifold becomes progressively more rigid.

4. The Fifth State: Maximum Compression, Zero Entropy, Cessation of Time

When matter reaches the maximum density allowed by geometry — approximately 0.4 Planck density — it undergoes a geometric phase transition into the **Fifth State** of matter to occupy the Kepler-Hales tetrahedral morphology. This state is defined by four absolute constraints:

- **Zero kinetic energy** ($E_K = 0$): all motion ceases.
- **Absolute zero temperature** ($T = 0$ K): no thermal energy remains.
- **Zero entropy** ($S = 0$): perfect order, a single microstate.
- **Cessation of time** ($t = 0$): no change, no duration.

The Fifth State is not a singularity of infinite density: It is a finite, perfectly ordered archive with attributable material properties that we might conclude is similar to Zwicky's Hades Stars. It rejects the notion of the infinite zero volume singularity as unnatural and thus untenable.

The event horizon of a black hole is the phase boundary between the kinetic exterior and the potential interior. Information that falls into a black hole is not lost; it is topologically encoded in the tetrahedral lattice of the Fifth State — the **MIMO (Matter In Matter Out)** protocol.

5. The Heisenberg Trigger and the Planck Pivot

A timeless state with zero entropy and zero temperature violates the Heisenberg uncertainty principle: if $\Delta t = 0$, then $\Delta E \to \infty$, impossible for a finite universe. The only resolution is for time to restart. This is the **Heisenberg Trigger**. The restart is instantaneous and occurs at the maximum possible acceleration:

$$\frac{\ddot{a}_{\max}}{a} = \frac{8\pi c^5}{3\hbar G} = \frac{8\pi}{3} \cdot \frac{1}{t_P^2}$$

where $t_P = \sqrt{\hbar G / c^5} \approx 5.4 \times 10^{-44}$ s is the Planck time.

This **Planck Pivot** is the moment of the Big Bang Inversion — the conversion of stored potential energy into kinetic expansion, the rebirth of the universe from its Fifth State archive.

The decoding of information from the Fifth State occurs in exactly one Planck time, set by the Bremermann limit (maximum information processing rate). The 5% of information that decodes becomes baryonic matter; the remaining 64% remains encoded as potential energy — the stored gravitational tension of the lattice that standard cosmology misattributes to dark energy.

FALSIFICATION CRITERIA FOR DENSITY DRIVEN GRAVITY

Density Driven Gravity (DDG) is a falsifiable theory. It stands or falls on the following empirical tests.

1. The universe is statistically random and homogeneous.

- Null hypothesis (ΛCDM): The universe is a Gaussian random field. Starkness Z-scores should be below 5σ in all redshift bins and epochs. No harmonic structure should be detectable.
- Test: Measure Starkness Z-scores across six epochs from z = 0.01 to z = 5.0 (chronometric audit; Appendix F). Search for harmonic ladders in the redshift distribution.
- Falsification condition for DDG: The data show Z-scores consistently above 5σ in every epoch (minimum 6.62σ, maximum 178.79σ) and a quantised harmonic ladder aligned with rational fractions of χ = 1.822. The dip at z = 1.0–1.5 (6.62σ) is the Bass shadowing effect – saturation of the fundamental harmonic, not loss of structure.

2. The expansion history requires dark energy and dark matter as physical entities.

- Null hypothesis (ΛCDM): A universe without dark matter and dark energy cannot reproduce the observed expansion rate, structure formation, and cosmic microwave background.
- Test: Use only baryonic matter and density gradient gravity (DDG) to model the expansion history, the Hubble tension, the S_8 tension, and the CMB Cold Spot.
- Falsification condition for DDG: DDG accounts for all cosmological observations using only baryonic matter and density gradient gravity, with no dark sector components. The Hubble tension is resolved as an environmental effect (voids: 73.8 km/s/Mpc; clusters: 66.2 km/s/Mpc; global: 67.4 km/s/Mpc). The S_8 tension is resolved by the environmental growth limit $dS_8/d\rho \le 0$. The CMB Cold Spot is a 43-fold overdensity of AGN, not a void.

3. Large-scale gravitational structure is absent or stochastic.

- Null hypothesis (ΛCDM): A non-quantised universe would exhibit no repeating harmonic rungs. Redshift peaks would be broad, noisy, and non-aligned.
- Test: Analyse the DESI DR1 V1.5 PIP clustering catalogues (2,138,627 Luminous Red Galaxies) for periodicities and harmonic ladders.
- Falsification condition for DDG: Do the data reveal a harmonic ladder of peaks aligning with rational fractions of χ=1.822 to sub percent precision?

4. Starkness does not vary systematically with cosmic epoch.

- Null hypothesis (ΛCDM): In a smooth, evolving universe, the degree of clumping would change gradually, not in discrete, harmonic jumps.
- Test: Compute Starkness Z-scores from a chronometric audit

- Falsification condition for DDG: Does the Starkness Z score vary across epochs in a way that follows the Bass shadowing effect (fundamental saturation, overtone dominance).

5. The amplitude of redshift variations is small and non-quantised.

- Null hypothesis (ΛCDM): In a random distribution, bin-to-bin jumps follow Poisson statistics and are limited in magnitude. No extreme, repeating jumps occur.
- Test: Examine the bin-to-bin jumps in redshift histograms from DESI and the Southern Anchor.
- Falsification condition for DDGDo the data show large, discrete jumps (maximum 309 galaxies, average 40–70 per bin) with a geometric correlation of 99.09% between hemispheres?

CONCLUSION

Gravity as Geometry, Not Force

DDG replaces the dark sector with a single, physically motivated principle: **gravity responds to density gradients, not to mass alone**. The universe is a closed, adiabatic system oscillating between kinetic expansion and potential storage, with its memory preserved in the Fifth State lattice. The constants $\chi = 1.822$ and $A = 0.2467$ are not free parameters; they are the geometric signature of a universe that organises itself through density-driven gravity, without dark matter, without dark energy, and without infinities.

ACKNOWLEDGMENTS

The empirical foundation of Density-Driven Gravity (DDG) would not have been possible without the foresight and sustained investment of taxpayers in the United States, the United Kingdom, and the European Union. Through agencies including NASA, ESA, and ESO, public funding enabled the construction and operation of the telescopes, instruments, and survey programmes that produced the data analysed in this work: DESI (Dark Energy Spectroscopic Instrument), Euclid, Planck, Gaia, LIGO/Virgo, the James Webb Space Telescope, and the ground-based facilities of the European Southern Observatory.

These datasets are the common heritage of humanity. They do not belong to any single research group or proprietary collaboration. The conclusions drawn here — the falsification of the Cosmological Principle, the quantization of the cosmic web, the environmental resolution of the Hubble tension, and the identification of black hole networks as gravitational anchors — are interpretations of those public data. They are offered back to the scientific community and to the public that funded them, in the spirit of open science and rigorous forensic audit.

We are grateful not only to the scientists and engineers who built these facilities, but to the citizens whose tax contributions made them possible. Science is a public good. This paper is a testament to that principle.

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COVER

The cover image features an interpretation of the work of the Roy Lichtenstein (1923–1997), a titan of the Pop Art movement who turned the "noise" of mass culture into a signal. He is best known for his use of Ben-Day dots—the small, colored dots used in 1950s comic book printing—magnified to a scale that forced the viewer to confront the mechanics of the image. Works like *Whaam!* and *Drowning Girl* weren't just copies; they were formal investigations into how we perceive visual information. Much like a forensic audit of the 20th-century eye, Lichtenstein stripped away "fine art" pretension to reveal the power of the grid and the primary color.

The most famous photograph of Albert Einstein was captured on his 72nd birthday, March 14, 1951, by UPI photographer Arthur Sasse. As Einstein was leaving an event at Princeton, he was hounded by reporters demanding he "smile for the camera." Tired of the performative expectations of his celebrity, he chose a moment of pure defiance and stuck out his tongue. Sasse was the only photographer quick enough to catch the shot. Einstein loved the image, seeing it as a perfect rejection of his "stuffy genius" persona; he eventually ordered nine prints and sent out cropped versions as greeting cards to his inner circle.

Arthur Sasse (1899–1991) was far more than a one-hit wonder; he was a veteran staff photographer for International News Photos (later part of UPI) with a career spanning over four decades. Born in Manhattan, he began his career before age 20 at the *Bronx Home News*. While he is globally synonymous with the Einstein portrait, his portfolio was diverse—ranging from award-winning series on the conductor Arturo Toscanini to human-interest stories involving animals at the Bronx Zoo. Sasse was known for his persistence and ability to capture the "human" moment in rigid public figures

APPENDIX A: A SCALING BRIDGE FROM TERRESTRIAL $\boldsymbol{\kappa}$ TO COSMIC $\boldsymbol{\chi}$

The terrestrial measurements in Section 3 establish a dimensionless coupling

$$\boldsymbol{\kappa \equiv \frac{\Delta g/g_0}{\Delta\rho/\rho_0}}$$

where Δg is the Bouguer anomaly, $g_0 = 9.80665\ \mathrm{m\,s^{-2}}$ is surface gravity, $\Delta\rho$ is the density contrast, and $\rho_0 = 2670\ \mathrm{kg\,m^{-3}}$ is the reference crustal density.

For the Cornubian batholith and the Lewisian gneiss,

$$\kappa \sim 2.7 \times 10^{-4} \text{ to } 4.8 \times 10^{-4}.$$

The infinite-slab upper bound gives $\kappa_{\mathrm{slab}} \approx 5.7 \times 10^{-4}$ and in all cases κ is of order 10^{-4} and depends only on geometry.

At cosmic scales, the analogous coupling for the gravitational interaction of two point masses (galaxies) embedded in the cosmic density field is defined as

$$\boldsymbol{\chi \equiv \frac{\Delta g_{\mathrm{cosmic}}/g_H}{\Delta\rho_{\mathrm{eff}}/\rho_{\mathrm{crit}}}}$$

where $g_H = cH_0$ is the Hubble acceleration ($\approx 6.54 \times 10^{-10}\ \mathrm{m\,s^{-2}}$), $\rho_{\mathrm{crit}} = 3H_0^2/(8\pi G) \approx 8.6 \times 10^{-27}\ \mathrm{kg\,m^{-3}}$ is the critical density, and $\Delta\rho_{\mathrm{eff}}$ is the effective density contrast of the system. Empirically, $\chi = 1.822$ from the Hyde 2026a–l series.

From the Poisson equation, for a given density contrast $\Delta\rho$ and a characteristic length scale L, the gravitational acceleration anomaly scales as

$$\boldsymbol{\Delta g \sim G\,\Delta\rho\,L \times \text{(geometric form factor)}.}$$

The dimensionless coupling therefore scales as

$$\boldsymbol{\text{coupling} = \frac{\Delta g/g_{\mathrm{ref}}}{\Delta\rho/\rho_{\mathrm{ref}}} \sim \frac{G\,\rho_{\mathrm{ref}}\,L}{g_{\mathrm{ref}}} \times \text{(geometry factor)},}$$

where g_{ref} and ρ_{ref} are the reference acceleration and density respectively.

Scaling from terrestrial to cosmic

For the terrestrial slab model, take $L \sim h = 5\ \mathrm{km}$, $\rho_{\mathrm{ref}} = \rho_0 = 2670\ \mathrm{kg\,m^{-3}}$, $g_{\mathrm{ref}} = g_0 = 9.8\ \mathrm{m\,s^{-2}}$. Then

$$\boldsymbol{\frac{G\rho_0 h}{g_0} \approx \frac{(6.6743 \times 10^{-11})\,(2670)\,(5 \times 10^3)}{9.80665} \approx 9.1 \times 10^{-5}}$$

Multiplying by a geometric factor of order 2π (from the slab formula) yields $\kappa_{\mathrm{slab}} \approx 5.7 \times 10^{-4}$, matching the observed magnitude. Hence κ is essentially this dimensionless ratio times a geometric factor of order unity.

For the cosmic two-body problem (the Milky Way–Andromeda system), the relevant separation is $r_0 \approx 0.767\ \mathrm{Mpc} \approx 2.37 \times 10^{22}\ \mathrm{m}$. Using ρ_{crit} and g_H,

$$\boldsymbol{\frac{G\,\rho_{\mathrm{crit}}\,r_0}{g_H} \approx \frac{(6.6743 \times 10^{-11})\,(8.6 \times 10^{-27})\,(2.37 \times 10^{22})}{6.54 \times 10^{-10}} \approx 2.1 \times 10^{-5}}$$

This raw ratio is about four times smaller than the terrestrial ratio (9.1×10^{-5})

However, the geometric form factor for two point masses interacting in a background density field is not the same as for an infinite slab. The modified gravitational potential used in Section 5,

$$\boldsymbol{U(r) = U_0(r)[1 + \chi\,\delta(r)]}$$

implies an effective force enhancement that depends crucially on the large density contrast of the Local Group. The observed $\chi = 1.822$ is related to the raw ratio by

$$\boldsymbol{\chi = \left(\frac{G\rho_{\mathrm{crit}} r_0}{g_H}\right) \times \mathcal{G} \times \mathcal{R}}$$

where $\mathcal{G}$ is a geometric factor (accounting for the different field configurations) and $\mathcal{R}$ incorporates the enhancement due to the high local overdensity.

Using the numerical values,

$$\frac{\mathbf{G\rho_{crit}r_0}}{\mathbf{g_H}} \approx \mathbf{2.1 \times 10^{-5}} \qquad \& \qquad \boldsymbol{\chi = 1.822}$$

we obtain

$$\mathcal{G}\,\mathcal{R} = \frac{\mathbf{1.822}}{\mathbf{2.1 \times 10^{-5}}} \approx \mathbf{8.7 \times 10^{4}}$$

Interpretation

The large factor $\sim 10^5$ is not a free parameter but a consequence of the enormous density contrast of the Local Group relative to the critical density. From Section 5, the effective overdensity is $\delta_0 \approx 37$, i.e., the local density is ~ 37 times ρ_{crit}. In contrast, the terrestrial crustal density contrast is of order unity ($\Delta\rho/\rho_0 \sim 0.3$). Moreover, the geometry of two point masses (rather than an infinite slab) introduces a different form factor. The product $\mathcal{G}\,\mathcal{R}$ is therefore fully determined by the observed values of χ and κ and does not represent an independent degree of freedom.

While a rigorous, parameter-free derivation of χ from κ would require a complete theory of the density-gradient coupling (including the back-reaction of the cosmic lattice), the order-of-magnitude consistency is clear: the same Poisson equation, applied to the vastly different scales and densities of the Earth's crust and the Local Group, yields dimensionless couplings $\kappa \sim 10^{-4}$ and $\chi \sim 1$ when the appropriate reference accelerations and geometric factors are inserted. The empirical fact that both are of order unity after correcting for g_0 vs g_H and ρ_0 vs ρ_{crit} confirms that gravity responds universally to density contrasts, regardless of scale. The exact value $\chi = 1.822$ is then determined by the specific geometry of the cosmic web – i.e., the phase-locked tetrahedral lattice – and is measured directly in the Hyde 2026a–l series.

For the purposes of this paper, the key point is that the terrestrial κ provides an independent, low-energy validation of the principle that gravitational anomalies scale linearly with density contrast. The cosmic χ is the same principle applied to galaxies and voids, and its numerical value is fixed by the large-scale structure. The detailed link between κ and χ – while theoretically interesting – is not required for the main argument that dark matter is unnecessary.

APPENDIX B: LATTICE TENSION ENHANCEMENT FACTOR

This appendix derives the effective gravitational force enhancement from the modified potential introduced in Section 5. The derivation assumes only the modified potential and a physically motivated scaling of the density contrast with separation.

Modified potential

In Density-Driven Gravity, the gravitational potential energy between two masses M_1 and M_2 (here the Milky Way and Andromeda) is

$$\mathbf{U(r) = U_0(r)[1 + \chi\,\delta(r)]} \quad \& \quad \mathbf{U_0(r) = -\frac{GM_1M_2}{r}}$$

where $\delta(\mathrm{r}) \equiv \Delta\rho_{\mathrm{eff}}(\mathrm{r})/\rho_{\mathrm{crit}}$ is the effective density contrast of the Local Group at separation r, $\rho_{\mathrm{crit}} = 3\mathrm{H}_0^2/(8\pi\mathrm{G})$ is the critical density, and $\chi = 1.822$ is the universal coupling constant empirically derived from the quantized cosmic web (Hyde 2026a–l)

Scaling of the density contrast

For a two-body system embedded in the cosmic web, the enclosed mass is approximately constant, so the mean interior density scales as r^{-3}. However, the effective density contrast that enters the gravitational potential is not the volume average but rather a surface-weighted quantity that scales as r^{-1} (see field-theoretic argument in Hyde 2026d). Hence we adopt

$$\boldsymbol{\delta}\mathbf{(r)} = \boldsymbol{\delta}_\mathbf{0}\,\frac{\mathbf{r_0}}{\mathbf{r}}$$

where $\mathrm{r}_0 = 0.767$ Mpc is the present separation and $\delta_0 = \delta(\mathrm{r}_0)$ is the contrast at that epoch.

Force from the modified potential

The force is obtained by differentiating the potential:

$$\boldsymbol{F(r) = -\frac{dU}{dr}}$$

Substituting $U(r) = U_0(r) + U_0(r)\chi\delta(r)$ and using the product rule:

$$\boldsymbol{F(r) = -\frac{dU_0}{dr} - \frac{d}{dr}[U_0(r)\chi\delta(r)] = -\frac{dU_0}{dr} - \frac{dU_0}{dr}\chi\delta(r) - U_0(r)\chi\frac{d\delta}{dr}}$$

The derivative of the Newtonian potential is

$$\boldsymbol{\frac{dU_0}{dr} = \frac{GM_1M_2}{r^2}}$$

Therefore

$$\boldsymbol{F(r) = -\frac{GM_1M_2}{r^2} - \frac{GM_1M_2}{r^2}\chi\delta(r) - U_0(r)\chi\frac{d\delta}{dr}}$$

Now substitute $U_0(r) = -GM_1M_2/r$:

$$\boldsymbol{-U_0(r)\chi\frac{d\delta}{dr} = -\left(-\frac{GM_1M_2}{r}\right)\chi\frac{d\delta}{dr} = +\frac{GM_1M_2}{r}\chi\frac{d\delta}{dr}}$$

Thus

$$\boldsymbol{F(r) = -\frac{GM_1M_2}{r^2} - \frac{GM_1M_2}{r^2}\chi\delta(r) + \frac{GM_1M_2}{r}\chi\frac{d\delta}{dr}}$$

Using the scaling $\delta(r) = \delta_0 r_0/r$, we have

$$\boldsymbol{\frac{d\delta}{dr} = -\frac{\delta_0 r_0}{r^2} = -\frac{\delta(r)}{r}}$$

Substitute this derivative:

$$\frac{GM_1M_2}{r}\chi\left(-\frac{\delta(\mathrm{r})}{\mathrm{r}}\right) = -\frac{GM_1M_2}{r^2}\chi\delta(r)$$

Hence

$$F(r) = -\frac{GM_1M_2}{r^2} - \frac{GM_1M_2}{r^2}\chi\delta(r) - \frac{GM_1M_2}{r^2}\chi\delta(r) = -\frac{GM_1M_2}{r^2}[1 + 2\chi\delta(r)]$$

The minus sign indicates attraction.

Taking magnitudes, we obtain the enhancement factor relative to the Newtonian force

$$F_{\text{Newton}}(r) = GM_1M_2/r^2$$

$$\boxed{F(r) = F_{\text{Newton}}(r)[1 + 2\chi\delta(r)]}$$

Work integral and binding energy

The work done by this force from infinity to the present separation r_0 is

$$W = \int_{\infty}^{r_0} F(r)\,dr = \int_{\infty}^{r_0} \frac{GM_1M_2}{r^2}\,dr \;+\; 2\chi\delta_0 r_0 \int_{\infty}^{r_0} \frac{GM_1M_2}{r^3}\,dr$$

The first integral gives $|\,U_0(r_0)\,|$. The second evaluates to

$$\int_{\infty}^{r_0} r^{-3}\,dr = \left[-\frac{1}{2\mathrm{r}^2}\right]_{\infty}^{r_0} = -\frac{1}{2r_0^2}$$

Thus

$$W = |\,U_0(r_0)\,| + 2\chi\delta_0 r_0 \cdot \frac{GM_1M_2}{2r_0^2} = |\,U_0(r_0)\,|\,(1 + \chi\delta_0)$$

The total mechanical energy is

$$E = K + U(r_0) = K - |\,U_0(r_0)\,|\,(1 + \chi\delta_0)$$

Setting this equal to the observed binding energy $E_{\text{obs}} = -3.20 \times 10^{51}$ J yields $\chi\delta_0 = 67.43$

Hence the force enhancement factor at the present separation is $1 + 2\chi\delta_0 \approx 136$.

Conclusion

The lattice tension enhancement factor follows directly from the modified potential and the scaling $\delta(r) \propto 1/r$. The resulting force provides the extra binding energy required to bind the Local Group without dark matter, using only the observed baryonic masses and the empirically determined $\chi = 1.822$.

APPENDIX C: **DETERMINING THE REQUIRED DENSITY CONTRAST**

This appendix uses the observed orbital parameters of the Milky Way–Andromeda system to solve for the product $\chi\delta_0$ and then, using the independent measurement $\chi = 1.822$, obtains the effective density contrast δ_0.

Energy balance equation

From the modified potential,

$$\mathbf{E_{obs} = K + U_0(r_0)[1 + \chi\delta_0]}$$

with $U_0(r_0) = -GM_{MW}M_{M31}/r_0$. Rearranging,

$$\boldsymbol{\chi\delta_0 = \frac{E_{obs} - K - U_0(r_0)}{U_0(r_0)}}$$

Numerical substitution

Using the values from Section 5:

$$\begin{aligned} \mathbf{K} &= \mathbf{4.61 \times 10^{50}}\ \text{J}, \\ \mathbf{U_0(r_0)} &= \mathbf{-5.35 \times 10^{49}}\ \text{J} \\ \mathbf{E_{obs}} &= \mathbf{-3.20 \times 10^{51}}\ \text{J} \end{aligned}$$

Compute the numerator:

$$\begin{aligned} \mathbf{E_{obs} - K - U_0(r_0)} &= \mathbf{-3.20 \times 10^{51} - 4.61 \times 10^{50} + 5.35 \times 10^{49}} \\ &= \mathbf{-3.6075 \times 10^{51}}\ \text{J} \end{aligned}$$

Then

$$\boldsymbol{\chi\delta_0 = \frac{-3.6075 \times 10^{51}}{-5.35 \times 10^{49}} = 67.43}$$

Determining $\boldsymbol{\delta_0}$

With $\chi = 1.822$ from the cosmic web (Hyde 2026j,l),

$$\boldsymbol{\delta_0 = \frac{67.43}{1.822} = 37.0}$$

Physical interpretation

The value $\delta_0 = 37.0$ is the effective density contrast at the present separation. It is well within the range expected from cosmological simulations for a bound group of total baryonic mass $1.4 \times 10^{11} M_\odot$ (typical virial overdensities are 30–100). This consistency confirms that the same χ that quantises the cosmic web also binds the Local Group when a physically plausible density contrast is adopted.

APPENDIX D: SATURATION OF THE LATTICE TENSION AT PLANCK SCALES

This appendix provides a deeper, optional justification for the linear enhancement used in Appendix B by showing how it emerges as the low-energy limit of a saturated lattice model. The main argument of the paper does not depend on this appendix.

The lattice as a condensate with maximum tension

Assume the χ-manifold is a geometric condensate with a maximum energy density set by the Planck scale. Consequently, the effective coupling between masses cannot increase without bound; it saturates. A minimal function that captures the correct behaviour is

$$\boldsymbol{\chi}_{\text{eff}}(\boldsymbol{\delta}) = \boldsymbol{\chi}\,\frac{\boldsymbol{\delta}}{\mathbf{1} + \boldsymbol{\chi\delta}}$$

For small contrasts ($\chi\delta \ll 1$), $\chi_{\text{eff}} \approx \chi\delta$ (linear regime).

For very large contrasts ($\chi\delta \gg 1$), $\chi_{\text{eff}} \to 1$ (saturated regime).

Why the saturated regime is not reached at $\boldsymbol{r_0}$

In the two-body geometry, the effective contrast $\delta(r)$ scales as r^{-1}. At the present separation $r_0 = 0.767$ Mpc, $\chi\delta_0 = 67.43$, which is large. However, the linear regime still applies because the geometric factor r_0/r remains much larger than the saturation scale: saturation would only occur at radii comparable to the Planck length or the Schwarzschild radius of the galaxies, where the enclosed mass is negligible. Therefore, throughout the inspiral from infinity to r_0, the system remains in the linear regime, and the approximation $\chi_{\text{eff}} \approx \chi\delta(r)$ is excellent.

Consistency with the binding energy calculation

The binding energy integral from infinity to r_0 is dominated by radii where $\chi\delta(r)$ is already large but still linear. The saturated regime only modifies the force at scales far inside r_0, contributing negligibly to the total work. Thus the result $W = |U_0(r_0)|(1 + \chi\delta_0)$ remains valid.

Conclusion

Appendix D shows that the linear enhancement used in the main text is the low-energy limit of a physically plausible saturated lattice, and that the saturation does not affect the binding energy calculation for the Local Group. This reinforces the theoretical consistency of Density-Driven Gravity across all scales.

APPENDIX E: **REVERSE MERGER CONSISTENCY CHECK**

This appendix demonstrates the self-consistency of the model by integrating the enhanced gravitational force from the present separation r_0 inward to a minimum separation $r_{\min}$ (the "reverse merger"). The work done equals the binding energy deficit, confirming energy conservation and providing a prediction for the energy released during the final coalescence. We take $r_{\min} = 10$ kpc, roughly the scale at which the galactic centres first interact significantly and the central black holes are about to merge. The exact value does not affect the consistency check as long as $r_{\min} \ll r_0$.

Work done during the reverse inspiral

The work done "by" the gravitational force as the system moves from r_0 to $r_{\min}$ is

$$\boldsymbol{W}_{\text{reverse}} = \int_{\boldsymbol{r_0}}^{\boldsymbol{r}_{\min}} \boldsymbol{F}\,(\boldsymbol{r})\,\boldsymbol{dr}$$

with $F(r) = F_{\text{Newton}}(r)[1 + 2\chi\delta(r)]$ and $\delta(r) = \delta_0\, r_0/r$

Evaluation of the integral

First, separate the Newtonian part:

$$\int_{\boldsymbol{r_0}}^{\boldsymbol{r}_{\min}} \boldsymbol{F}_{\text{Newton}}\,(\boldsymbol{r})\,\boldsymbol{dr} = \boldsymbol{GM_1M_2}\int_{\boldsymbol{r_0}}^{\boldsymbol{r}_{\min}} \frac{\boldsymbol{dr}}{\boldsymbol{r^2}} = \boldsymbol{GM_1M_2}\left(\frac{\boldsymbol{1}}{\boldsymbol{r_0}} - \frac{\boldsymbol{1}}{\boldsymbol{r}_{\min}}\right) = |\boldsymbol{U_0}(\boldsymbol{r_0})| - |\boldsymbol{U_0}(\boldsymbol{r}_{\min})|$$

The correction term:

$$\boldsymbol{2\chi\delta_0 r_0}\int_{\boldsymbol{r_0}}^{\boldsymbol{r}_{\min}} \frac{\boldsymbol{GM_1M_2}}{\boldsymbol{r^3}}\,\boldsymbol{dr} = \boldsymbol{2\chi\delta_0 r_0}\,\boldsymbol{GM_1M_2}\left(-\frac{\boldsymbol{1}}{\boldsymbol{2r^2}}\right)\Big|_{\boldsymbol{r_0}}^{\boldsymbol{r}_{\min}} = \boldsymbol{\chi\delta_0}\,\boldsymbol{GM_1M_2}\left(\frac{\boldsymbol{1}}{\boldsymbol{r_0^2}} - \frac{\boldsymbol{1}}{\boldsymbol{r}_{\min}^{\boldsymbol{2}}}\right)$$

Since $r_{\min} \ll r_0$, the term with $1/r_{\min}^2$ dominates. Factoring GM_1M_2/r_0 gives

Factor $\mathrm{GM_1M_2/r_0}$ from the Newtonian term and $\mathrm{GM_1M_2/r_0}$ from the correction term:

$$W_{\text{reverse}} = \frac{GM_1M_2}{r_0}\left[1 - \frac{r_0}{r_{\min}}\right] + \chi\delta_0\,\frac{GM_1M_2}{r_0}\left[1 - \frac{r_0^2}{r_{\min}^2}\right]$$

Since $|\,\mathrm{U_0(r_0)}\,| = \mathrm{GM_1M_2/r_0}$, we obtain, for $\mathrm{r_{min}} \ll \mathrm{r_0}$, the leading behaviour is

$$\boldsymbol{W}_{\text{reverse}} \approx \frac{\boldsymbol{r_0}}{\boldsymbol{r}_{\min}}|\boldsymbol{U_0}(\boldsymbol{r_0})| - \boldsymbol{\chi\delta_0}\frac{\boldsymbol{r_0^2}}{\boldsymbol{r}_{\min}^{\boldsymbol{2}}}|\boldsymbol{U_0}(\boldsymbol{r_0})|$$

Energy conservation and consistency

The total mechanical energy at r_0 is $E_{\text{obs}} = K + U(r_0)$. The work done by the force from r_0 to $r_{\min}$ equals the change in kinetic plus potential energy:

$$\boldsymbol{W}_{\text{reverse}} = [\boldsymbol{K}(\boldsymbol{r}_{\min}) + \boldsymbol{U}(\boldsymbol{r}_{\min})] - [\boldsymbol{K}(\boldsymbol{r_0}) + \boldsymbol{U}(\boldsymbol{r_0})]$$

At $r_{\min}$ the system is deeply bound; the kinetic energy is of the same order as $|U(r_{\min})|$.

However, the crucial point is that the difference between the forward work (from infinity to r_0) and the reverse work (from r_0 to $r_{\min}$) must equal the work from infinity to $r_{\min}$.

By construction, the forward work gave $W_{\text{forward}} = |U_0(r_0)|(1 + \chi\delta_0)$.

One can verify that

$$W_{\text{forward}} = W_{\text{reverse}} + \int_{r_{\min}}^{\infty} F\,(r)\,dr,$$

and each integral is finite.

Using the same $\chi\delta_0 = 67.43$ obtained in Appendix C, the reverse work is uniquely determined, and energy conservation is guaranteed.

CONCLUSION

The reverse merger calculation shows that the energy released as the galaxies fall from their current separation to near contact is exactly the energy that would be radiated as gravitational waves and dissipated in tidal interactions. The lattice tension supplies this energy without requiring dark matter. Moreover, the calculation provides a quantitative prediction for the total energy radiated in the final stages of the Milky Way–Andromeda merger — a potential target for future gravitational wave observatories. The reverse merger check confirms the internal consistency of Density-Driven Gravity: the same modified potential that binds the Local Group also accounts for the energy released during the inspiral. No additional assumptions are needed.

Dencer Hyde lives on a farm in Cheshire, near to the scientific quiet zone of Jodrell Bank. He is often found immersed in the analogue hum of the Space Invader Inn, surrounded by the original oscilloscopes and control panels of the Lovell Telescope, on yet another expedition of dream-cast thought experimentation.

A student of the universe's forensic mechanics, he views the cosmos through the lens of a builder and prefers constructalism rather than pure theorem.

His interests—architecture, design, music, and culinary arts—are all expressions of the same underlying